

Diagnostic Imaging of Gastrointestinal Disorders in Cats and Dogs

Focus on Fibre-Related and Common Clinical Conditions

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Education

- DVM from FVM IPB (1997)
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Session Overview



Introduction & Clinical Relevance

1

Imaging Modalities — Principles & Indications

2

Normal GI Anatomy on Imaging

3

Fibre-Related GI Conditions ★ FOCUS

4

5 Common GI Clinical Conditions

6 Case-Based Discussion

7 Summary, Algorithm & Takeaway





Why Diagnostic Imaging Matters in GI Disease

?

Clinical Signs Overlap

Vomiting, diarrhoea,
weight loss, anorexia -
same presentation,
very different
diagnoses

→

Guide treatment decisions

Medical vs. surgical
management —
imaging provides the
evidence

✓

Non-invasive characterisation

Identify lesion
location, extent, and
tissue type without
exploratory surgery

↑

Monitor therapy & prognosis

Serial imaging
quantifies response to
treatment and
disease progression

*GI disorders rank among the top 5 reasons for
consultation in small animal clinics worldwide*



Learning Objectives

- 01 Select the most appropriate imaging modality based on suspected GI condition and clinical context
- 02 Recognise key radiographic and ultrasonographic features of common GI disorders in cats and dogs
- 03 Identify specific imaging characteristics of fibre-related conditions — megacolon, bezoars, and dietary fibre-related IBD
- 04 Apply a structured diagnostic imaging algorithm to GI cases in your daily clinical practice



IPB University
— Bogor Indonesia —



School of
Veterinary Medicine
and Biomedical Sciences



RSHP
Rumah sakit hewan pendidikan



AICVIM
Asian College of Veterinary Internal Medicine



BVLC
BOGOR
VETERINARY
LEARNING
CENTER



my vets
by Siti Zaenab



Four Modalities — When to Use Each



Radiography

First-line screening

Indicated for:

- Foreign bodies
- Obstruction
- Megacolon
- Free gas
- Mass effect

Limitation: Poor soft tissue
contrast

Ultrasonography

Primary GI modality

Indicated for:

- Wall layering
- Motility
- Lymph nodes
- Effusion
- Real-time

Limitation: Gas interference ·
Operator-dependent

CT / MRI

*Advanced staging &
planning*

Indicated for:

- Neoplasia staging
- Pre-surgical mapping
- Bowel viability

Limitation: GA required · Cost
· Availability

Endoscopy

Mucosal + biopsy

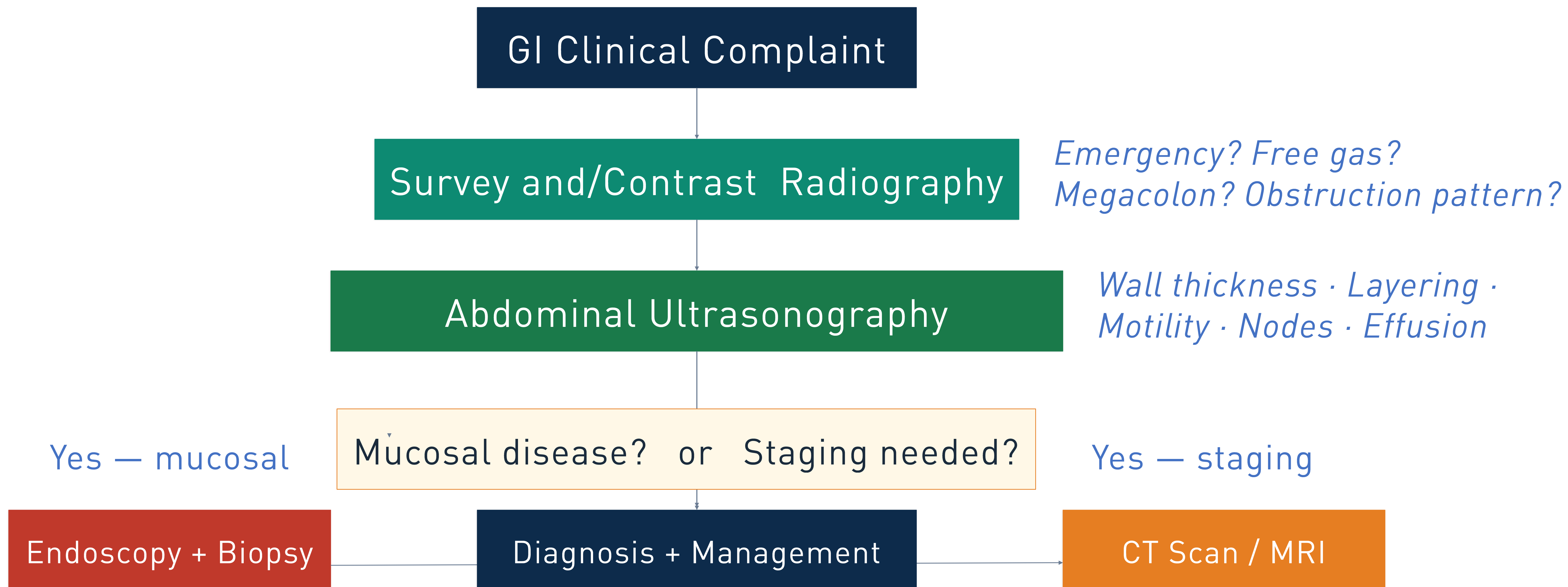
Indicated for:

- IBD confirmation
- Mucosal masses
- Haematemesis
source

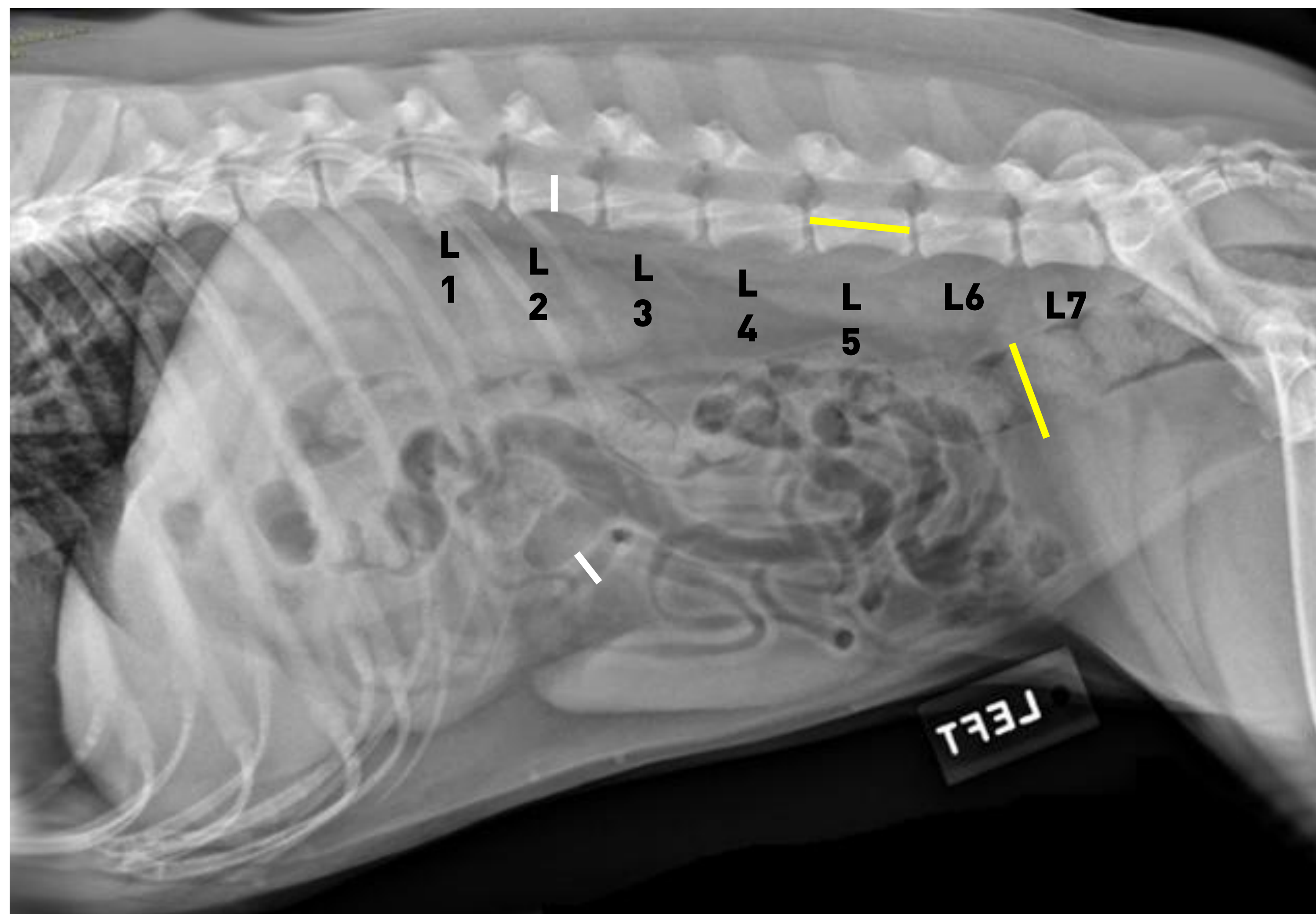
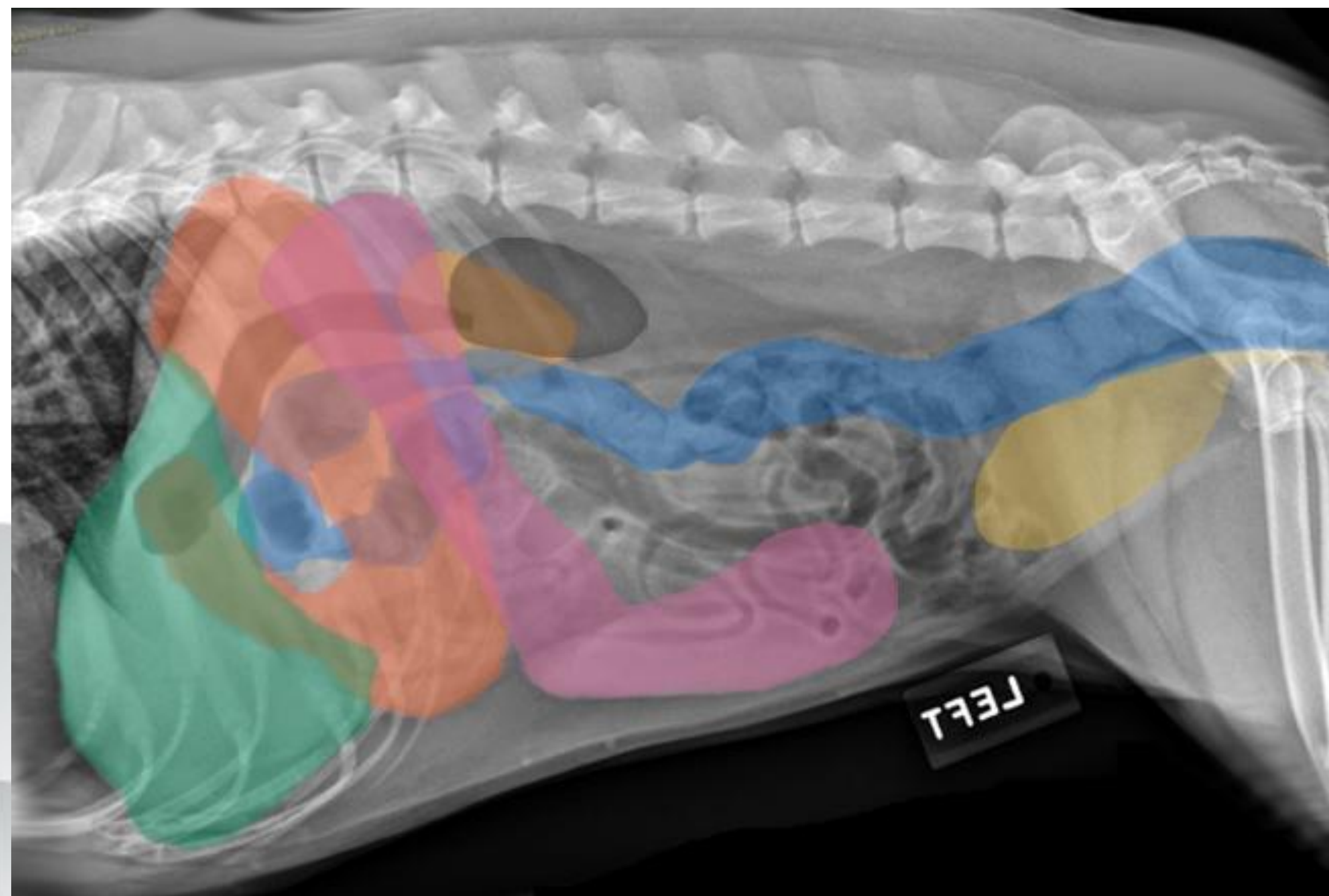
Limitation: Cannot assess
transmural disease



Modality Selection Algorithm



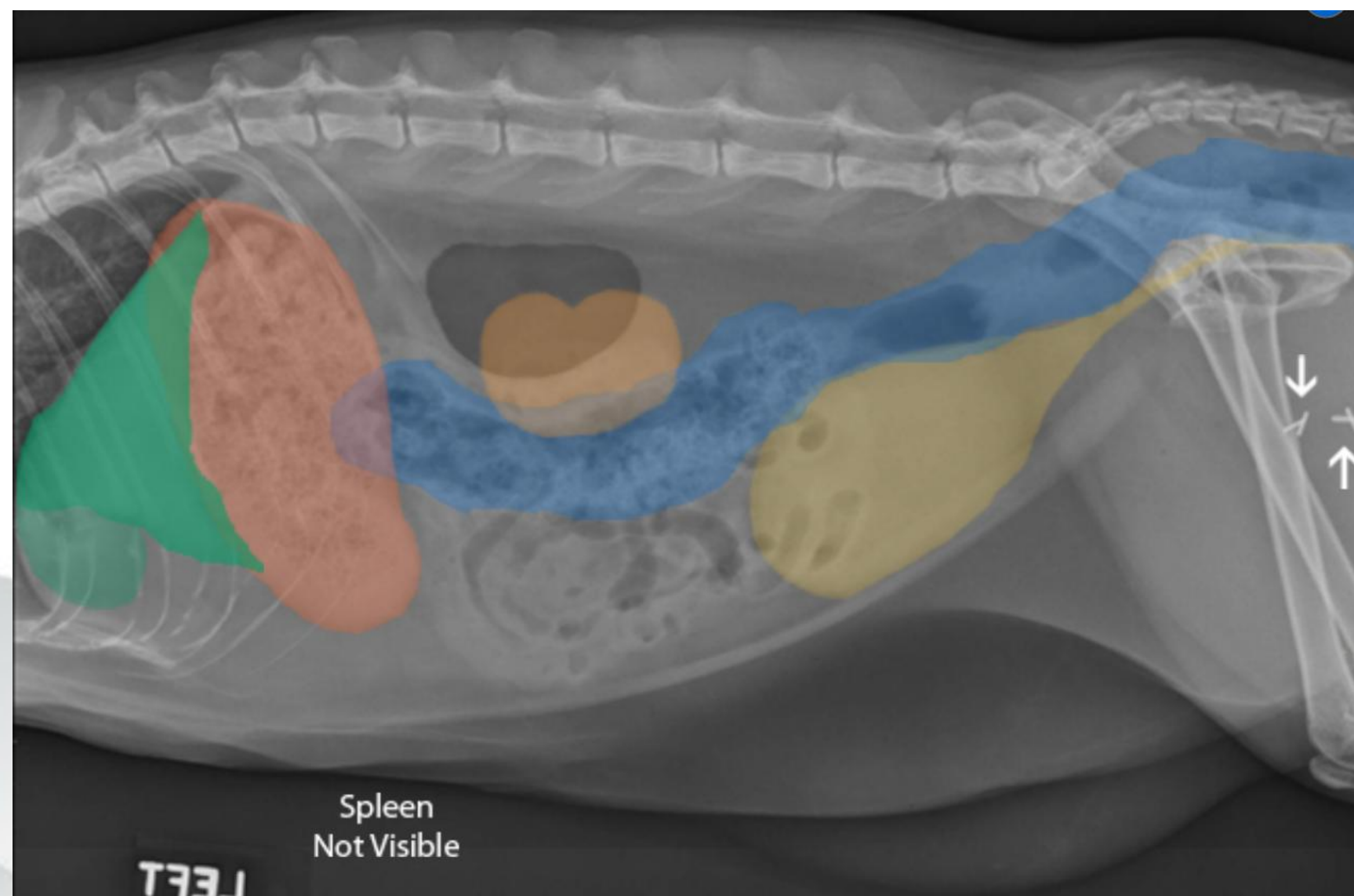
Canine Radiography



- Small intestinal loops **should not exceed 2x the height of the body of the second lumbar vertebra (L2).**
- The colon **should not exceed 1.5x the width of L5**



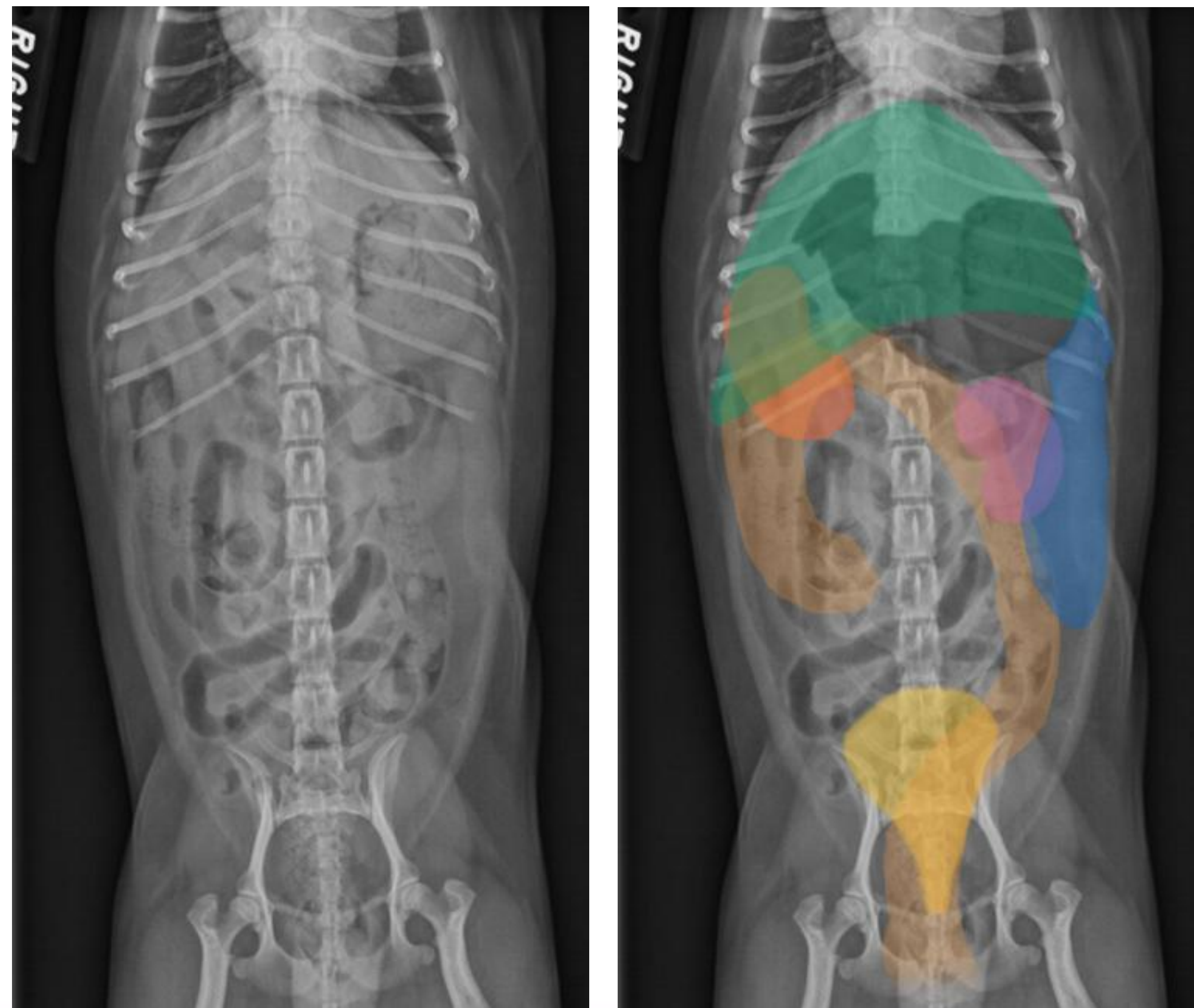
Feline Radiography



- **In cats, a more practical measurement is 12 mm maximum diameter for the small intestine.**
- Gas patterns on radiography: it is normal to see a small amount of gas within the stomach and colon.

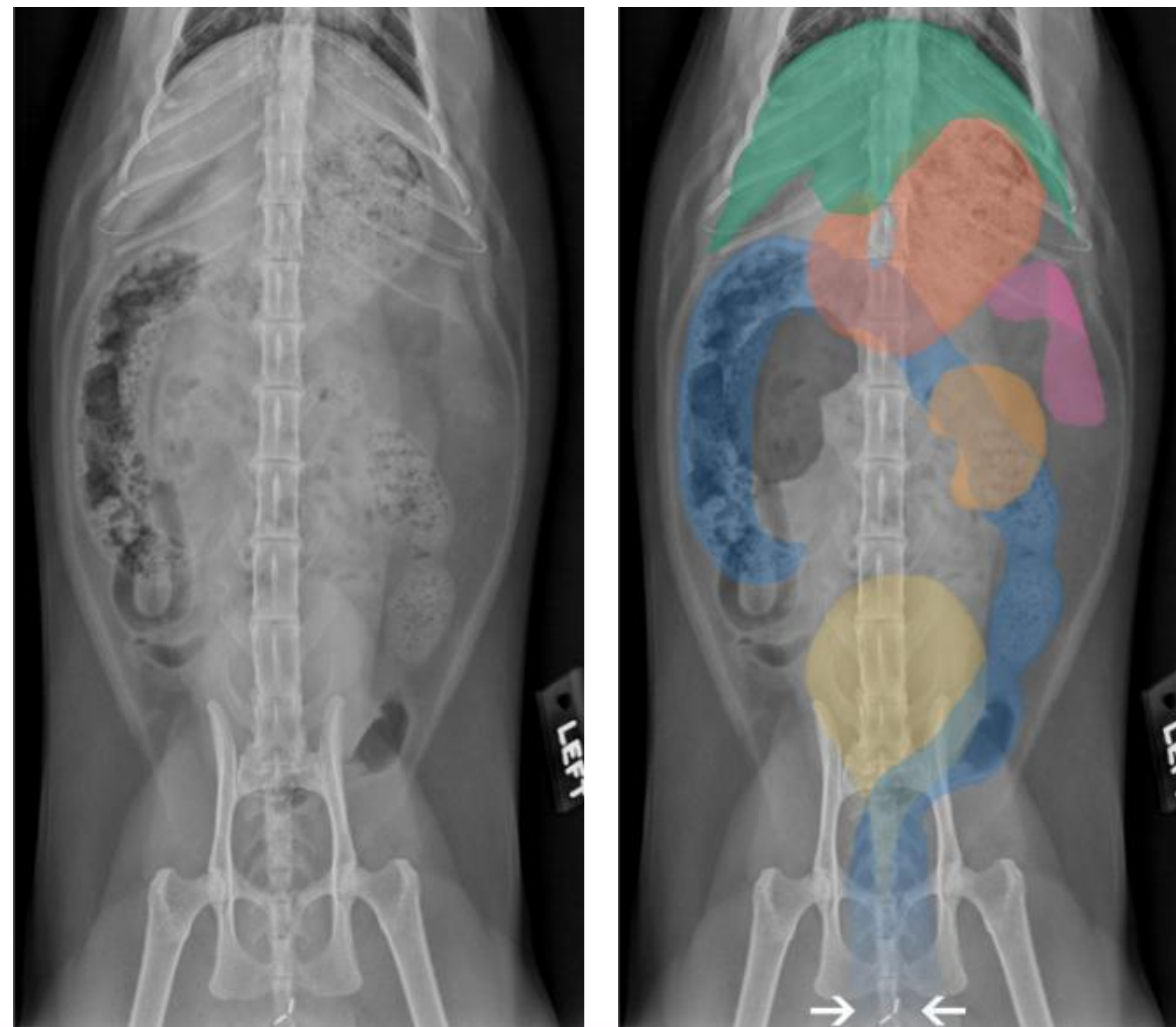


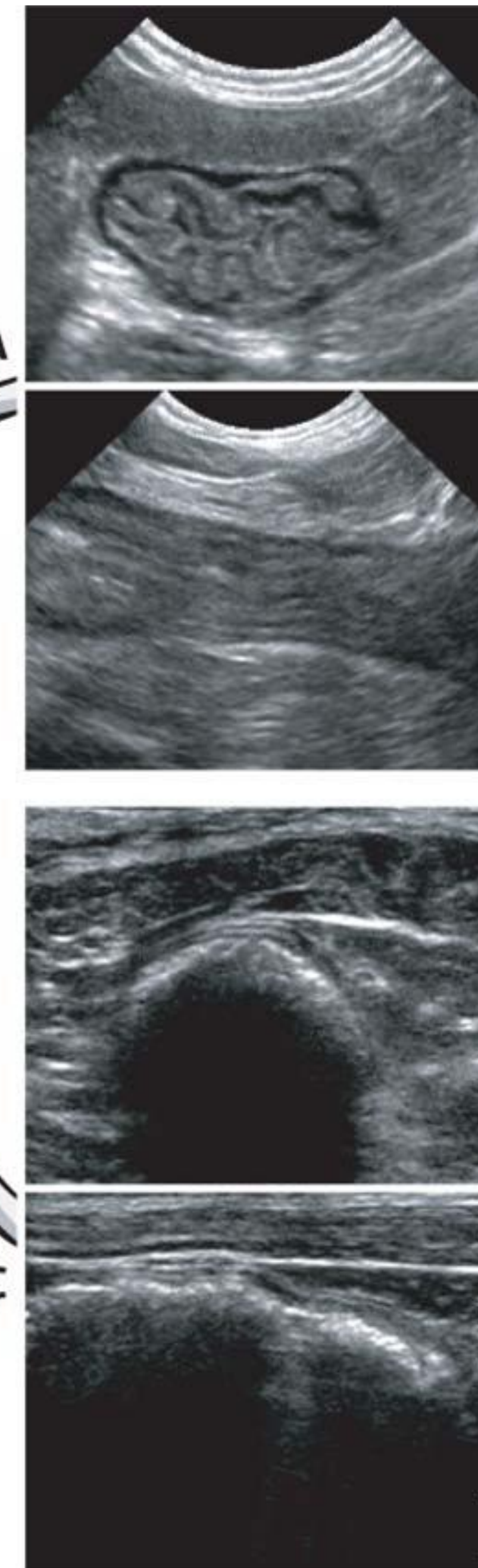
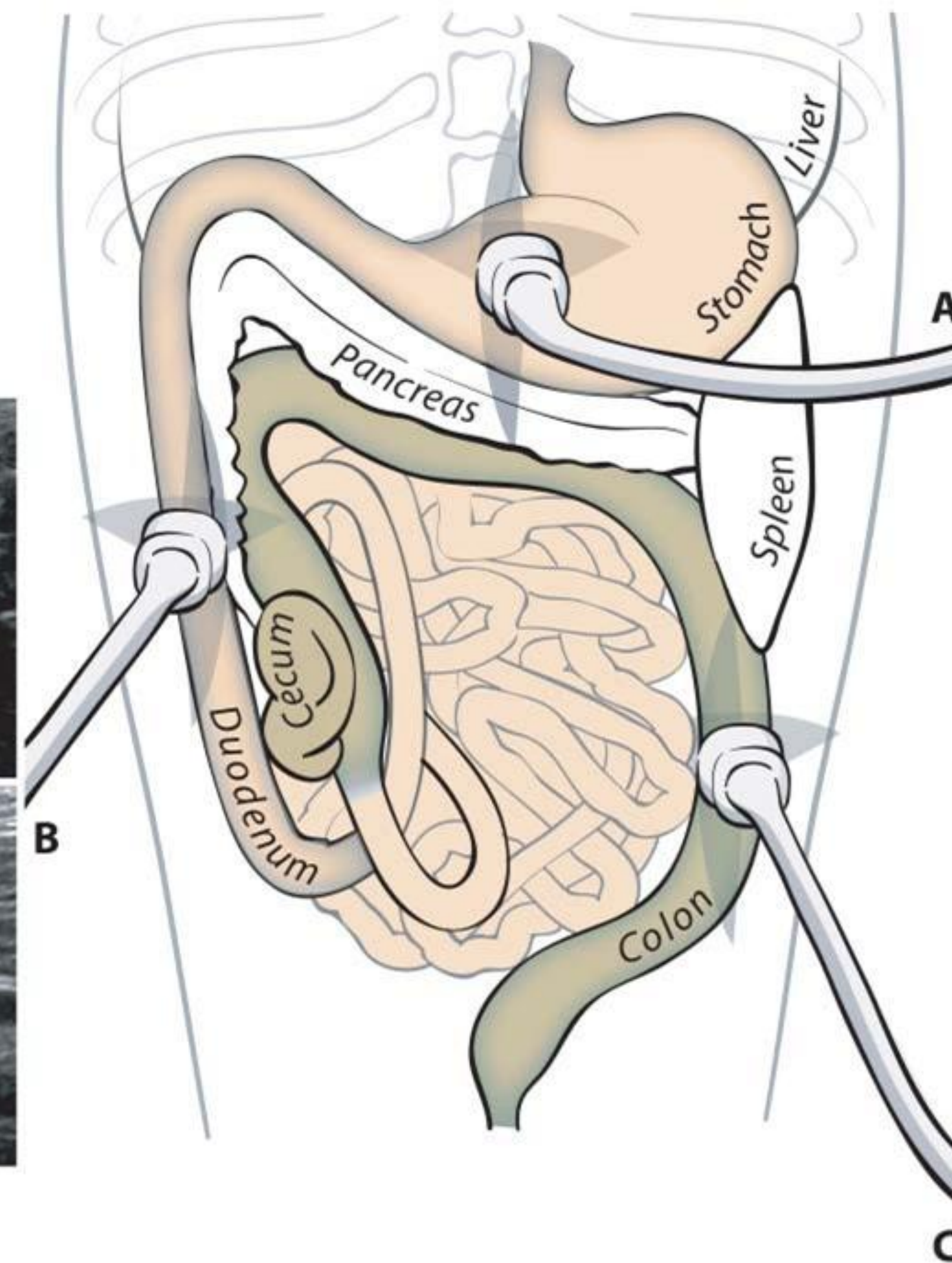
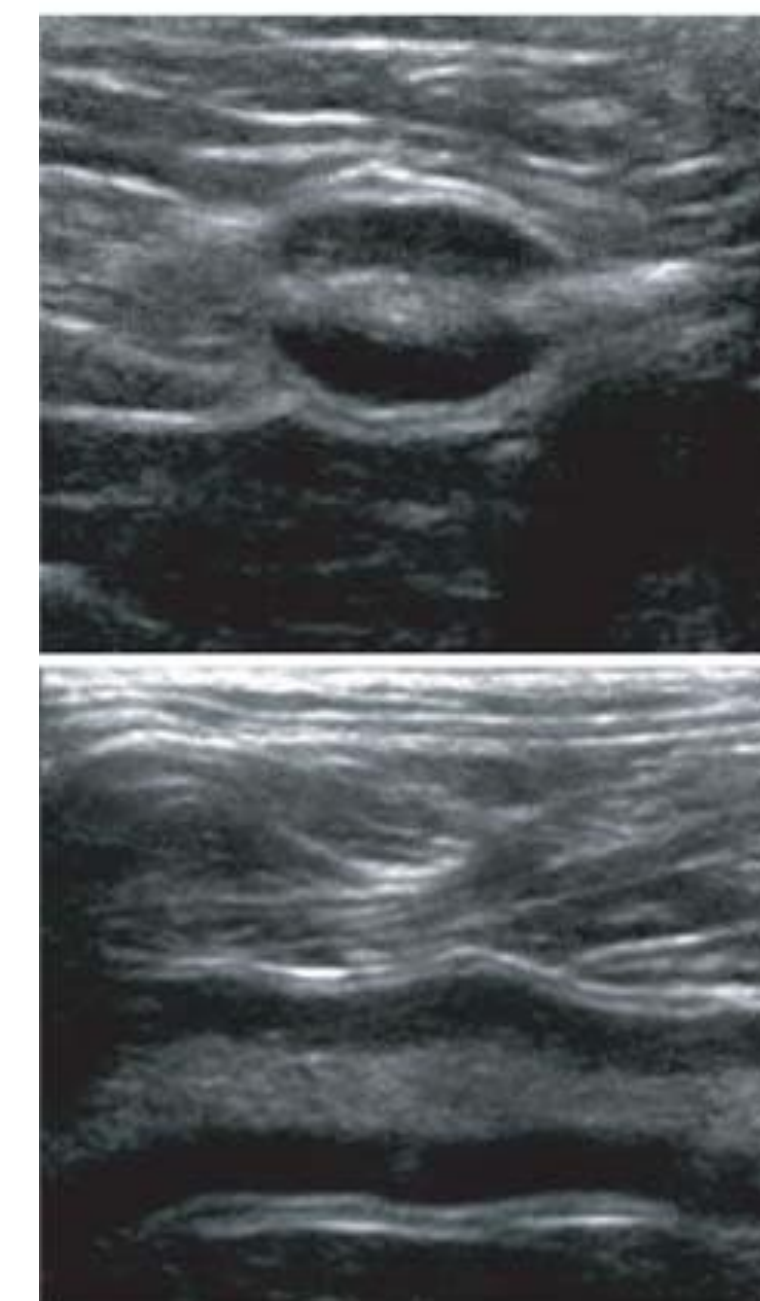
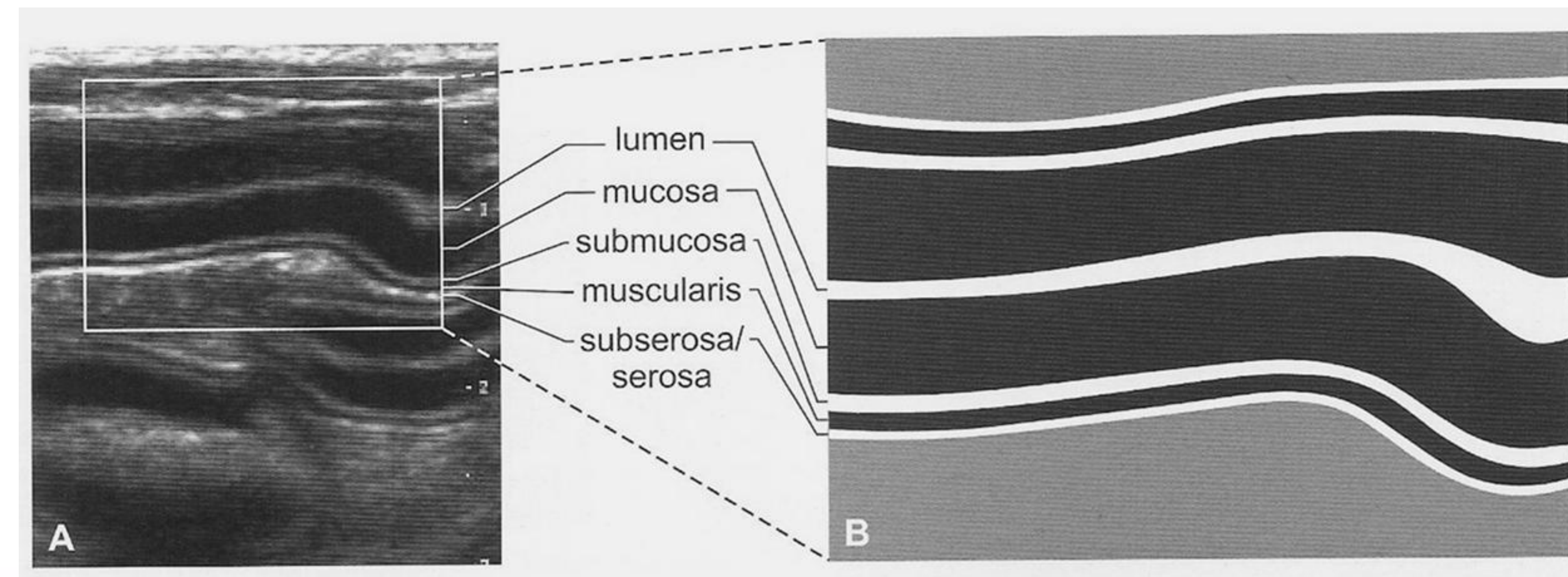
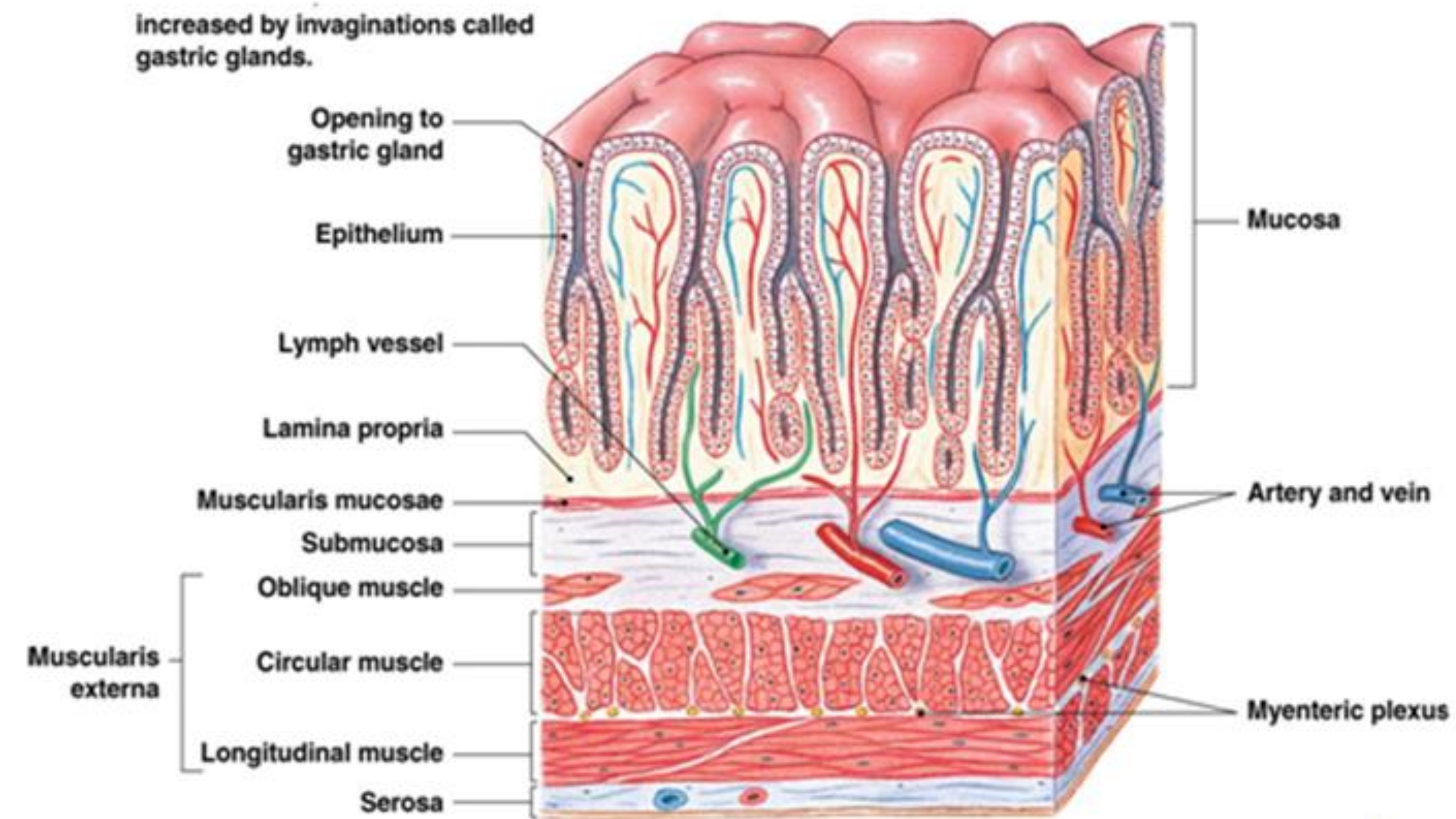
Canine Radiography



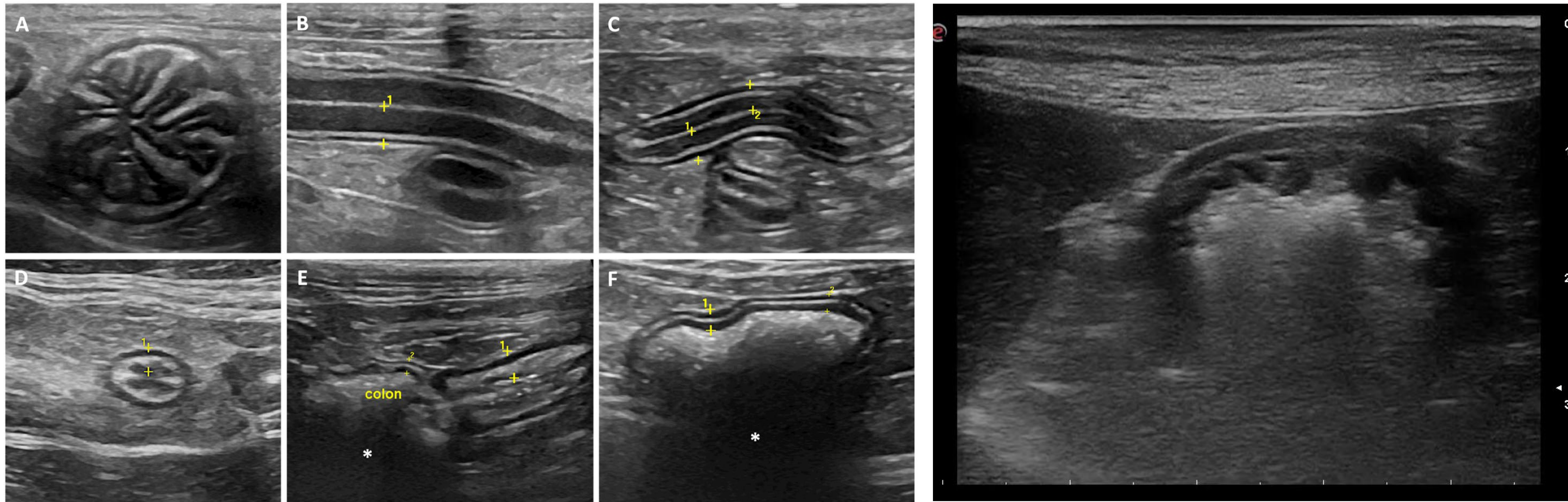
- Gas patterns on radiography: it is normal to see a small amount of gas within the stomach and colon.

Feline Radiography





Ultrasonographic images of the gastrointestinal segments



(A) empty stomach of a cat (rugular folds); (B) duodenum of a dog, and (C) jejunum of a cat (D) distal ileum of a cat; (E) Ileoceocolic region, (F) Colon, dog: gas filled, evidencing acoustic shadowing (*).



Barium vs. Iohexol — GI Contrast Study

Barium Sulfate (BaSO₄)

GOLD
STANDARD

USE	Mucosal detail, motility, suspected FB/bezoar, megaesophagus
DOSE	Dog 8–12 mL/kg · Cat 10–14 mL/kg (30–60% w/v suspension)
TIMING	Films at 0 · 15 · 30 · 60 min, then hourly until colon
GASTRIC T _{1/2}	Dog: empties <3–4 h · Cat: <4–6 h (liquid barium)
AVOID	Suspected perforation → causes granulomatous peritonitis

Iohexol (Water-soluble)

SAFE IF
PERFORATION

USE	Suspected perforation, pre-endoscopy, post-op leak check
DOSE	~2–3 mL/kg of 240–300 mgI/mL (non-ionic)
TIMING	Faster transit than barium; image at 15–30 min intervals
PROS	Safe if extravasation; resorbed from peritoneum
CONS	Weaker mucosal coating · dilutes rapidly · cost higher

KEY INDICATIONS

● Fibre / Bezoar

Trichobezoar (cat), phytobezoar — mottled/"honeycomb" filling defect as barium traps between fibres.

● Obstruction & FB

Radiolucent FB (plastic, cloth, rubber): filling defect. Linear FB → plicated loops + barium trail.

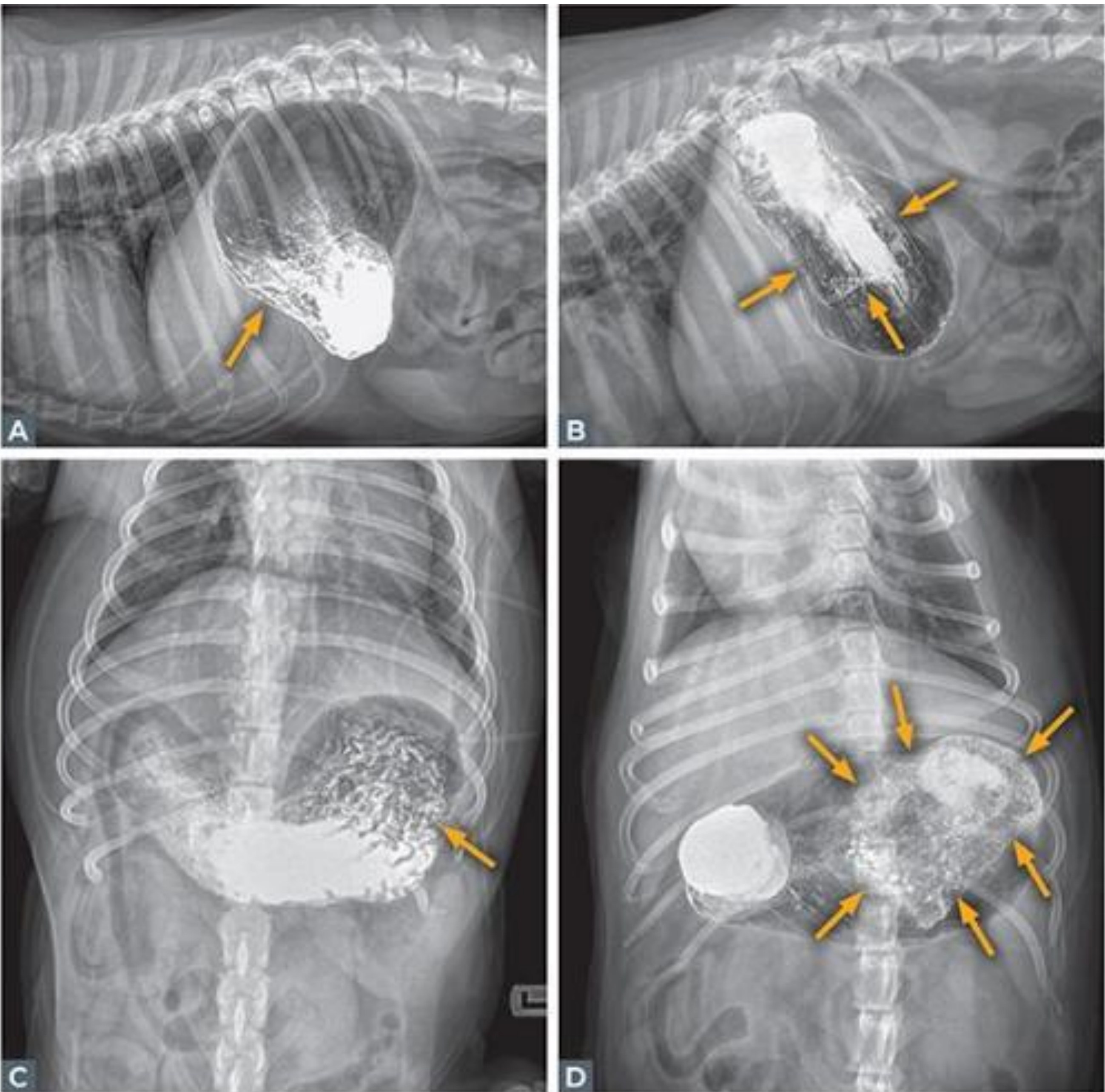
● Esophageal Disease

Megaesophagus, vascular ring anomaly, stricture, diverticulum. Use barium-mixed food for motility.

● Mucosal / Motility

Ulcer, mass, IBD (irregular mucosa), pyloric outflow obstruction, intussusception (coiled spring).

DON'T Use barium if perforation suspected · Skip contrast before endoscopy · Avoid in severe vomiting, aspiration risk, or hyperosmolar iodinated agents on dehydrated patients.



Dog with a gastric foreign body. The study shows a foreign body impregnated with barium resembling a sponge in the left lateral and ventrodorsal views (B, D). In the right lateral and dorsoventral views, the foreign body is not or barely visible due to the superimposition of the rugal folds.

Normal Reference Values on US & X-Ray Imaging



Structure	Dog — Normal	Cat — Normal	Key Modality
Small intestine (SI) wall	≤ 5 mm (duodenum ≤ 6 mm)	≤ 2.5 mm	Ultrasound
Stomach wall	≤ 7 mm (varies with distension)	≤ 4.5 mm	Ultrasound
SI diameter — max	≤ 2× height of L2 body	≤ 12 mm	Radiograph
Colon diameter — max	≤ 1.5× width of L5	≤ 1.5× width of L5	Radiograph
Normal SI layers (US)	5 layers: mucosa · submucosa · muscularis · subserosa · serosa		Ultrasound

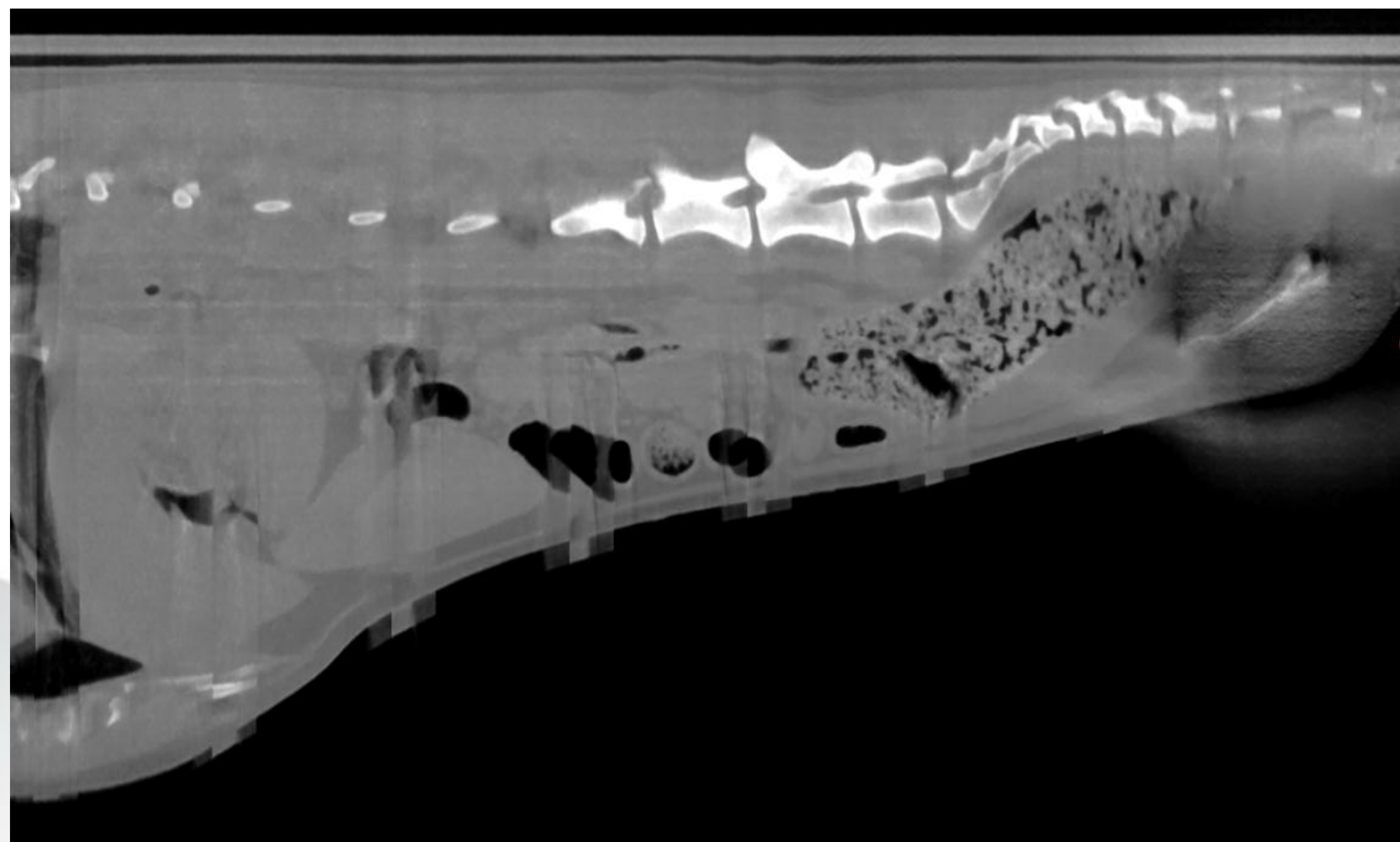


The 5-Layer Wall Architecture

Hyperechoic mucosa-lumen interface · Hypoechoic mucosa · Hyperechoic submucosa (brightest) · Hypoechoic muscularis propria · Hyperechoic serosa



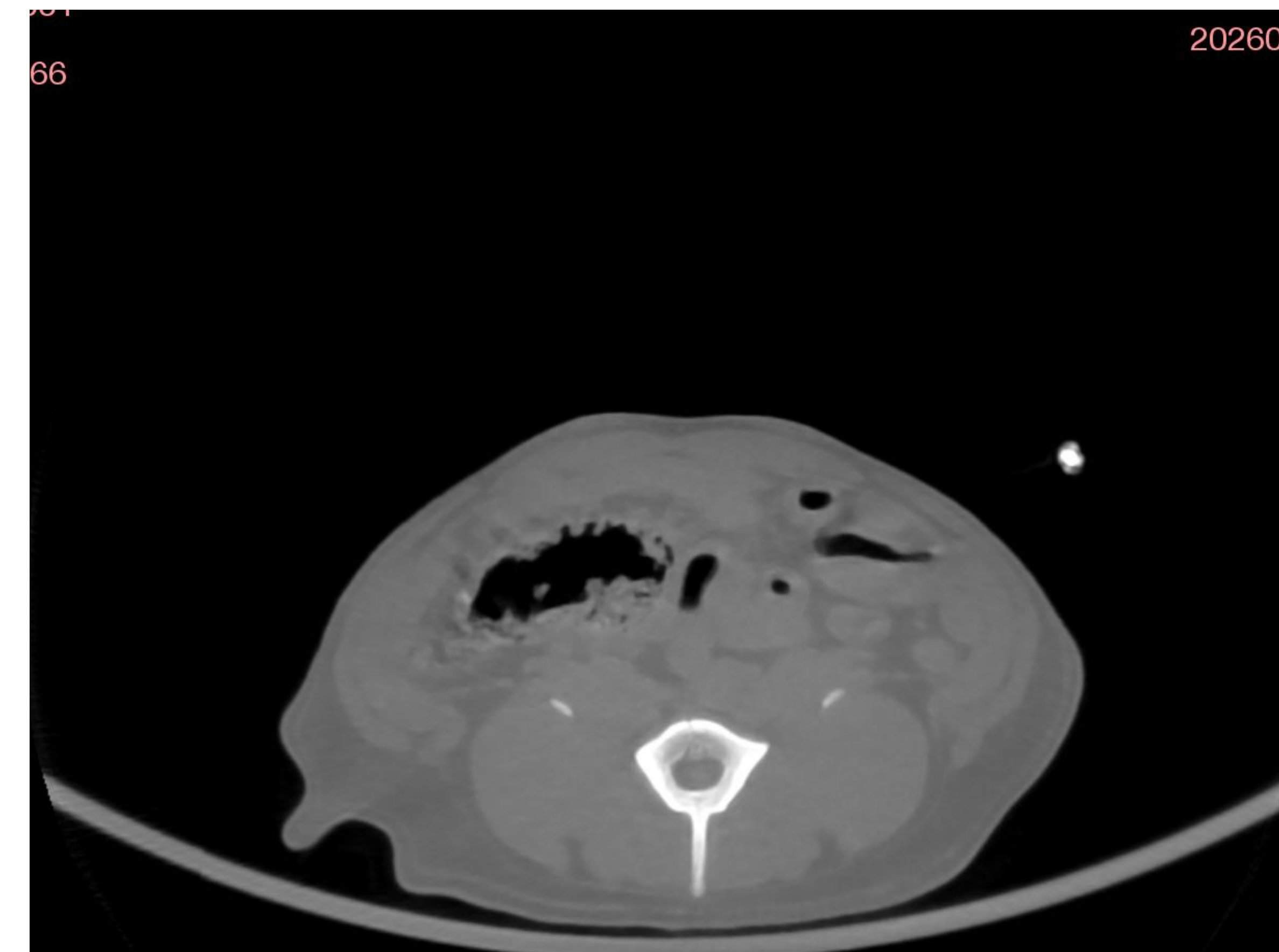
Canine Abdominal CT Scan



Sagittal



Coronal



Axial



6-Step Radiographic Read of the GI Tract



1 Size & Dilation

NORMAL Dog: SI loop $\leq 1.6 \times$ L5 body height. Cat: SI ≤ 12 mm.

RED FLAG Dilated loops (gas/fluid) + two populations of loops = mechanical obstruction.

2 Opacity

NORMAL Mixed gas & ingesta in stomach; small gas in SI; feces in colon.

RED FLAG Metallic/mineral FB · pneumoperitoneum (free gas) · loss of serosal detail.

3 Shape & Position

NORMAL Tubular, smoothly curving loops; gastric axis $\perp$ spine.

RED FLAG Plicated/bunched loops $\rightarrow$ linear FB. Displaced gastric axis $\rightarrow$ GDV.

4 Serosal Detail

NORMAL Crisp borders between loops, liver, kidneys.

RED FLAG "Ground glass" loss $\rightarrow$ peritoneal fluid, peritonitis, young/cachectic.

5 Luminal Contents

NORMAL Mixed gas/fluid/ingesta; variable throughout GI tract.

RED FLAG Pure fluid distension · "gravel sign" · geometric FB shapes.

6 2-View Rule

NORMAL Right lateral + VD (or DV) on every abdominal study.

RED FLAG If suspicious: add left lateral or standing horizontal beam for free gas.



6-Step USG Read of the GI Tract



1 Wall Thickness

NORMAL Dog: stomach ≤ 5 · duo ≤ 6 · jej ≤ 5 · colon ≤ 3 mm. Cat: SI ≤ 3 · colon ≤ 1.5 mm.

RED FLAG Diffuse thickening → IBD / lymphoma. Focal thickening → neoplasia / intussusception.

2 Wall Layering

NORMAL 5 distinct layers: mucosal line, mucosa, submucosa, muscularis, serosa.

RED FLAG Loss of layering → neoplasia (lymphoma, adenoca), severe inflammation, perforation.

3 Echogenicity

NORMAL Mucosa hypoechoic · submucosa & serosa hyperechoic · muscularis hypoechoic.

RED FLAG Hyperechoic mucosal speckles → lymphangiectasia. Thick hypoechoic muscularis (cat) → IBD/lymphoma.

4 Motility

NORMAL Stomach & duodenum 4–5/min · jejunum 1–3/min · colon infrequent.

RED FLAG Absent = ileus, obstruction, peritonitis. Hyperactive = enteritis or early obstruction.

5 Lumen Contents

NORMAL Small gas/fluid/mucus; colon with fecal acoustic shadow.

RED FLAG Fluid-filled loops with to-and-fro → obstruction. Shadowing FB · mass in lumen.

6 Surroundings

NORMAL Thin hypoechoic mesentery · small oval nodes · no free fluid.

RED FLAG Hyperechoic "steamy" mesentery → peritonitis / pancreatitis. Rounded LN · free fluid.



Common GI Conditions — Key Imaging Signs



Mechanical Obstruction / Foreign Body

XR: Gravel sign · Dilated SI loops proximal to FB · Radiopaque FB may be visible
US: Corrugation sign · Pendulous peristalsis · Linear FB → plication pattern (cats)

Intussusception

XR: Elongated soft tissue mass · Dilated SI loops proximal
US: 'Target sign' (transverse) · 'Sandwich sign' (longitudinal) · Doppler: assess bowel viability

Peritonitis / Effusion

XR: Free gas → perforation (emergency) · Generalised loss of detail
US: Anechoic or echogenic free fluid · Fibrin strands → exudative effusion · US-guided abdominocentesis

Gastric Dilatation-Volvulus (GDV)

XR: 'Double bubble' / compartmentalisation sign · Pylorus displaced dorsal-right on VD
US: Limited by gas · Evaluate hepatic/splenic perfusion · CT for wall viability

GI Neoplasia

XR: Mass effect · Displacement of bowel loops
US: Cats: diffuse SI thickening + muscularis prominence (lymphoma) · Dogs: focal mass, loss of layering · CT for staging

Pancreatitis + GI Involvement

XR: Sentinel loop (fixed, gas-filled duodenum)
US: Hypoechoic pancreas · Peripancreatic effusion · Rigid duodenum · CT for necrosis / complications



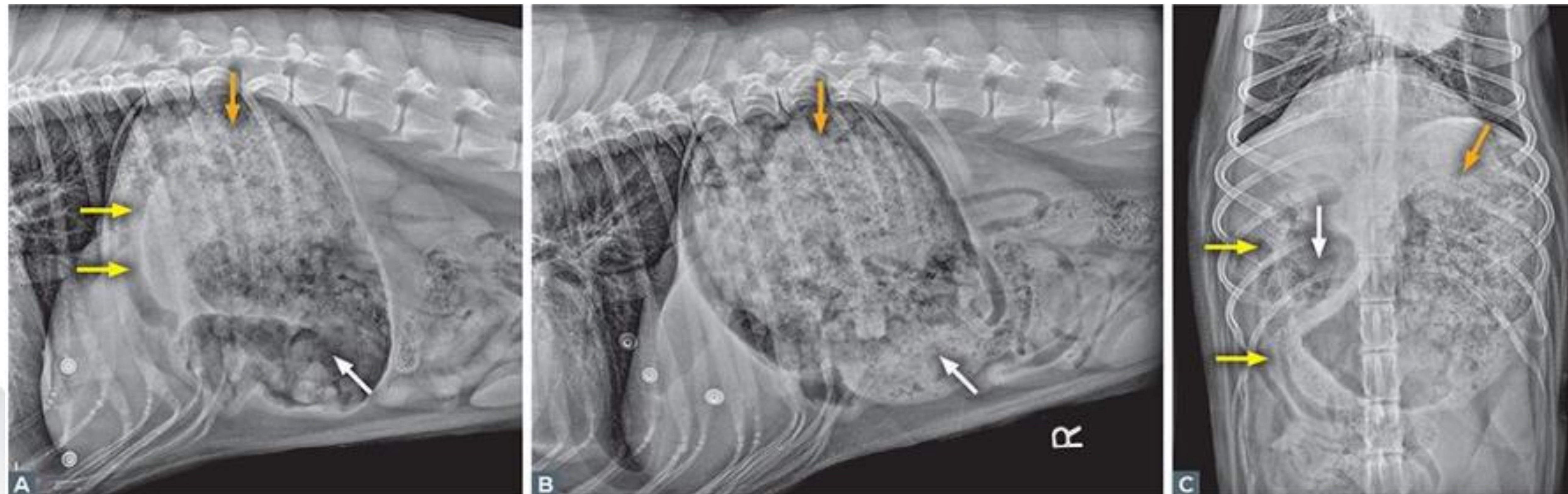
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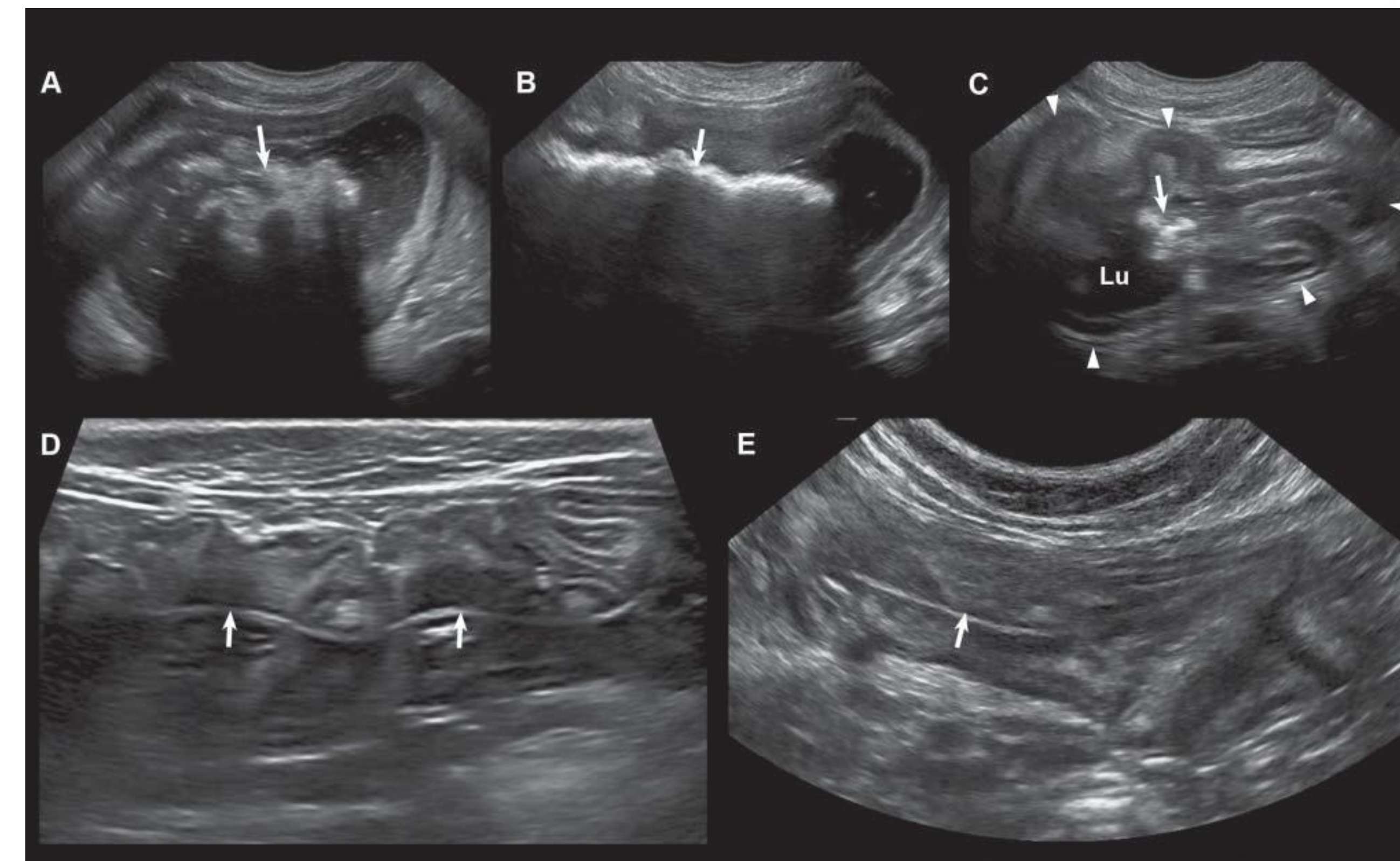


Food mixed with foreign materials in dog



The stomach has a normal position and markedly distended with a heterogeneous soft tissue and gas mixed ingesta that has a granular appearance. The duodenum contains gas (yellow arrows). The granular appearing content could represent food, foreign material, or a combination of food mixed with foreign material.

Linear foreign bodies in two dogs and one cat.



A-C: A 3-year-old English Bulldog. Foreign material (arrows) associated with strong shadowing is present in the stomach D-E: In a cat with a thin linear foreign body (arrows) extending into the small intestine, severely plicated bowel pattern is seen.



Intussusception

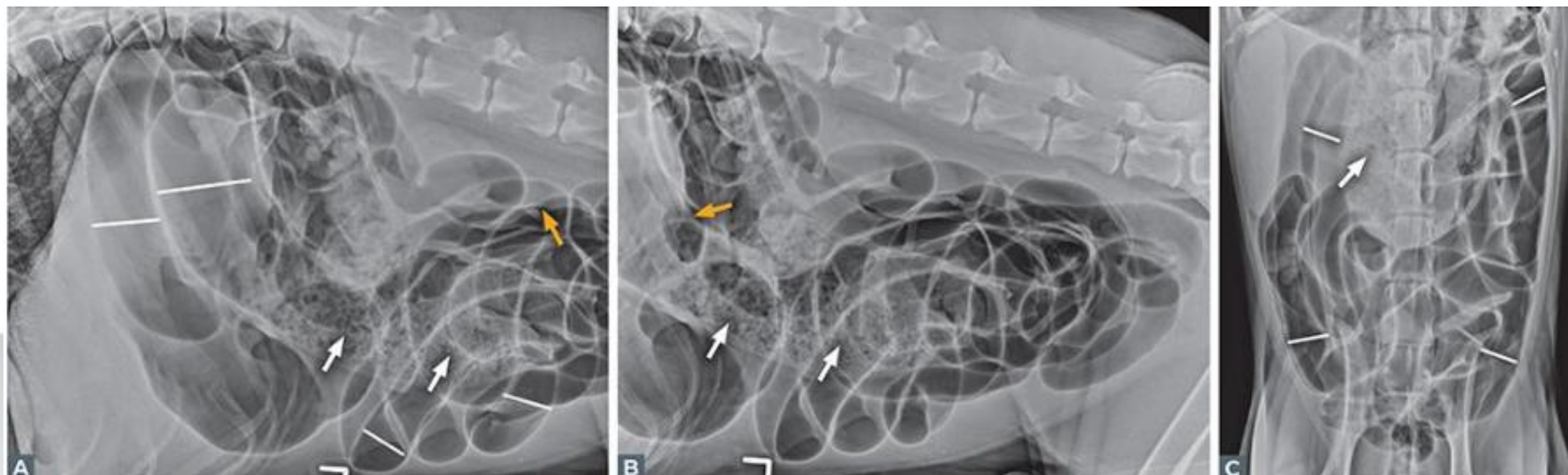
XR: Elongated soft tissue mass · Dilated SI loops proximal

US: 'Target sign' (transverse) · 'Sandwich sign' (longitudinal) ·

Doppler: assess bowel viability

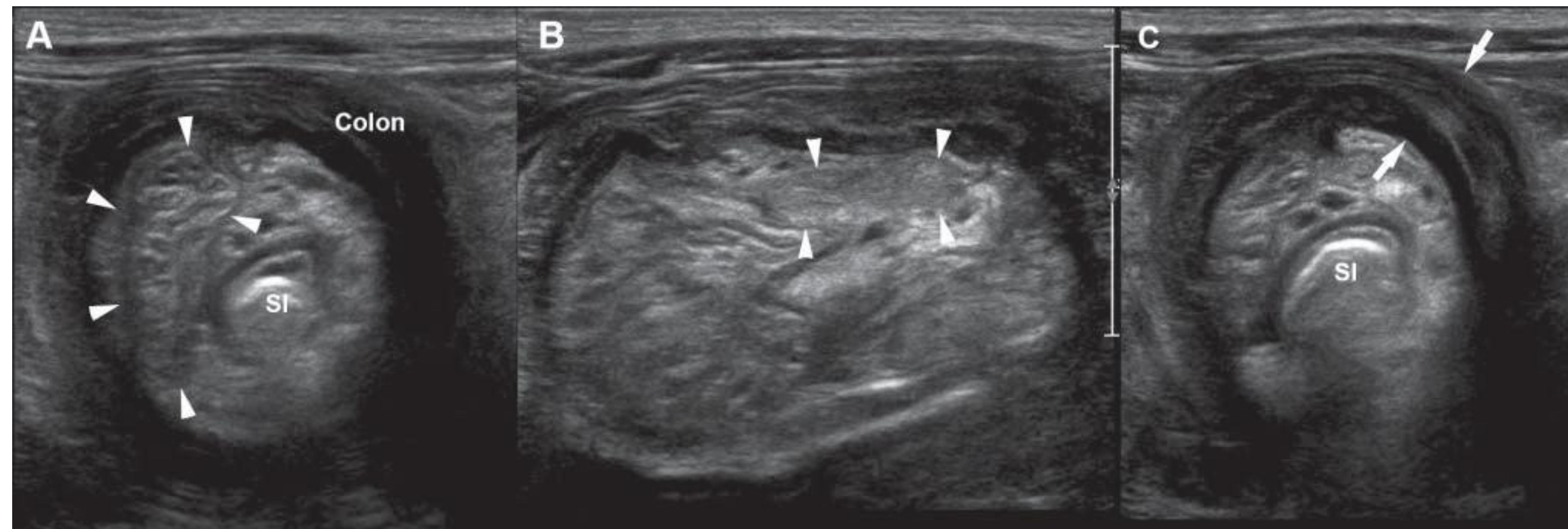


Jejunal or ileal obstruction, DDX Intussusception



4-year-old Great Dane presented with bloating, collapse, and hypovolemia. The entire small intestinal tract is severely dilated with gas (white bars show some of the intestines as an example, and the pattern is generalized). This pattern has differentials of end jejunal or ileal obstruction and mesenteric volvulus.

Jejuno-colic intussusception



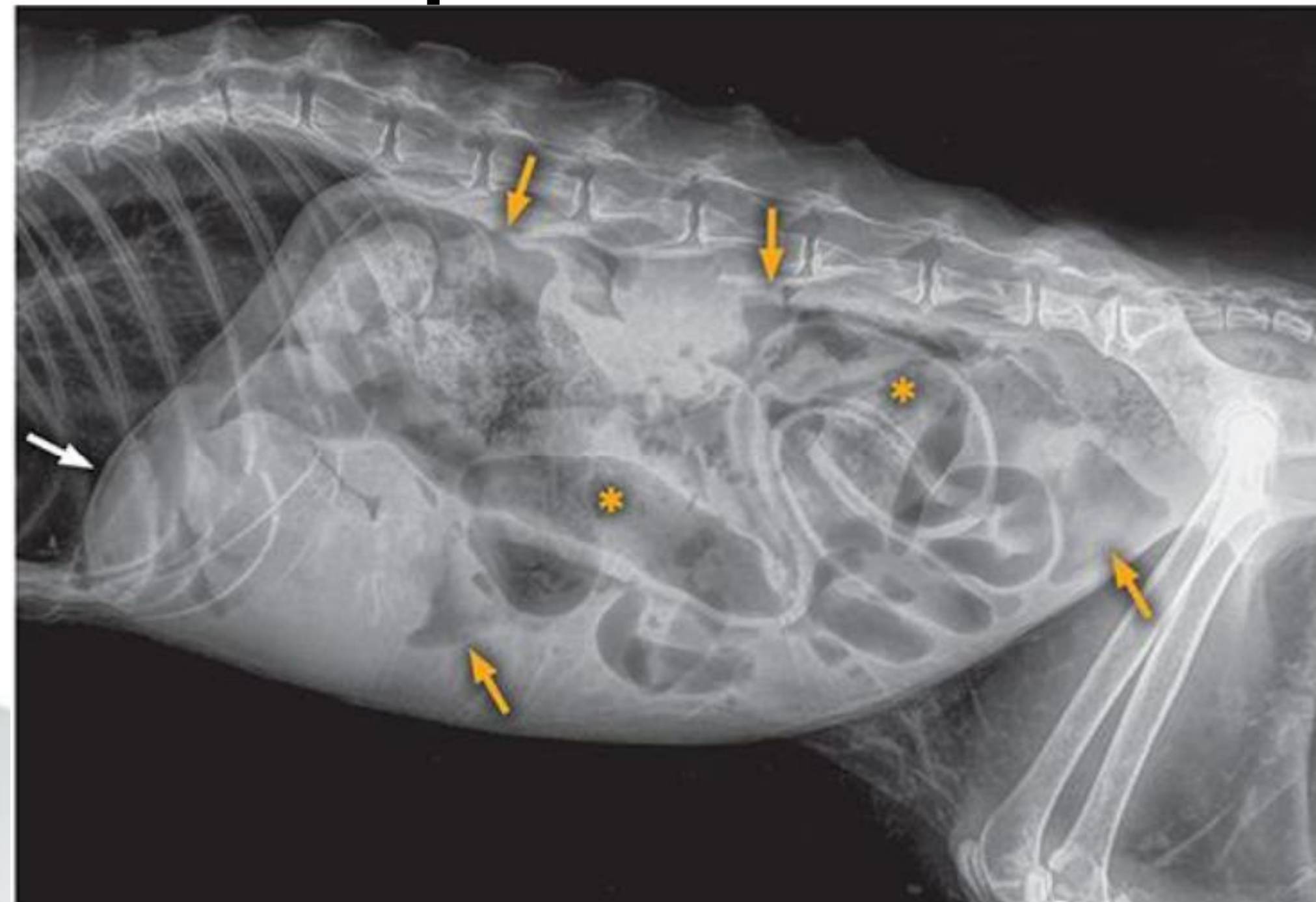
Transverse (A, C) and longitudinal (B) sonograms of a distal jejunocolonic intussusception detected in a 4-month-old Golden Retriever.



Peritonitis / Effusion

XR: Free gas → perforation (emergency) · Generalised loss of detail
US: Anechoic or echogenic free fluid · Fibrin strands → exudative effusion · US-guided abdominocentesis

Pneumoperitoneum & Peritonitis



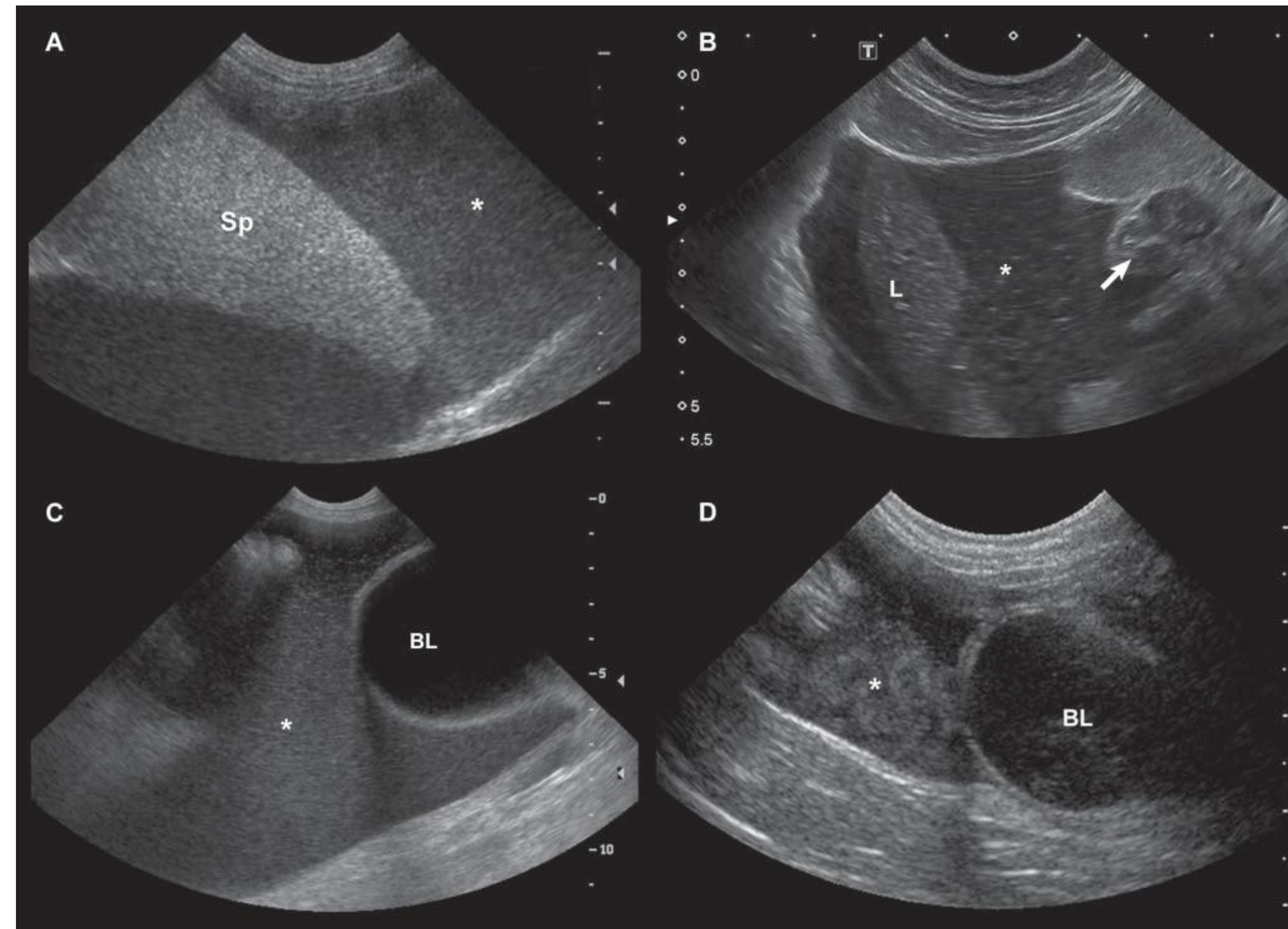
14 years Domestic longhair cat with cachexia. A few large, triangular gas bubbles are present throughout the peritoneal space (orange arrows).

Normal Radiography in Puppies



Two-month-old intact female French Bulldog presenting for vomiting. Notice the absence of serosal detail. This is a normal appearance for the patient's age.

Peritoneal effusions in dogs and cats



A: A large volume of particulate, echogenic, peritoneal effusion (*)

B: Peritoneal effusion (*), which presents hyperechoic speckles.

C: Dog chylous effusion (*) is typically of moderate to high echogenicity

D: A cat diagnosed with multifocal biliary carcinomas.

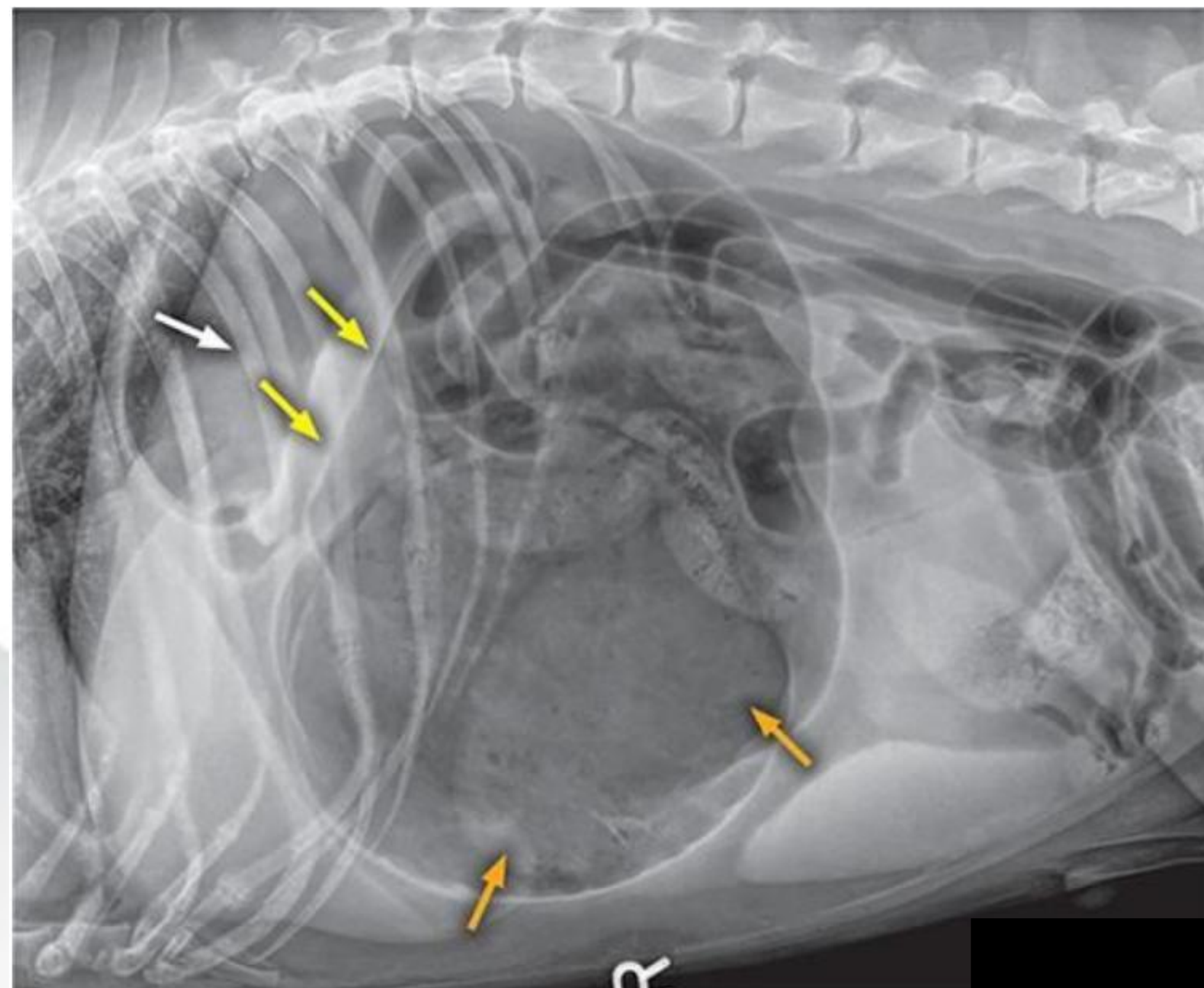


Pancreatitis + GI Involvement

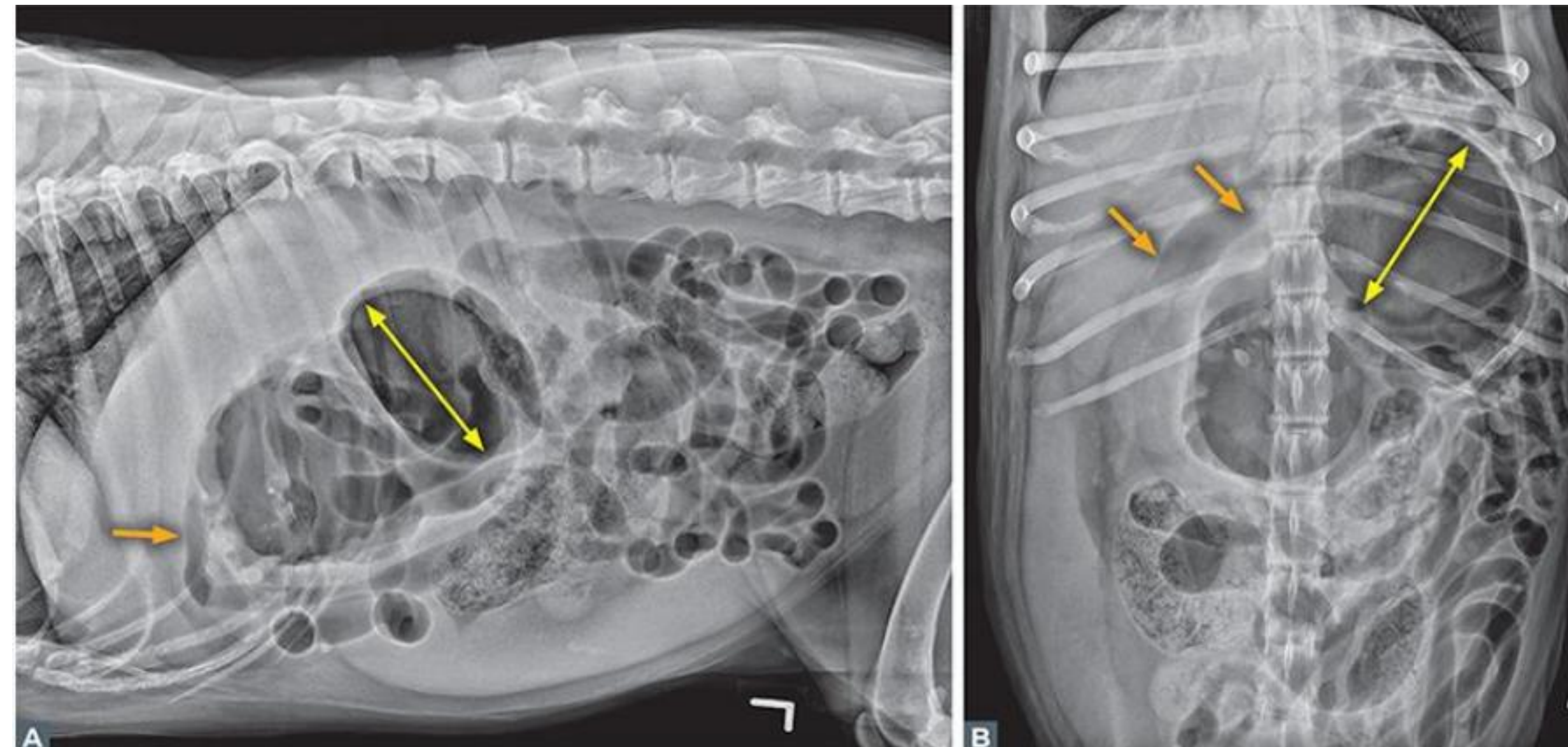
XR: Sentinel loop (fixed, gas-filled duodenum)

US: Hypoechoic pancreas · Peripancreatic effusion · Rigid duodenum · CT for necrosis / complications

Complete gastric dilation and volvulus



Partial volvulus of stomach

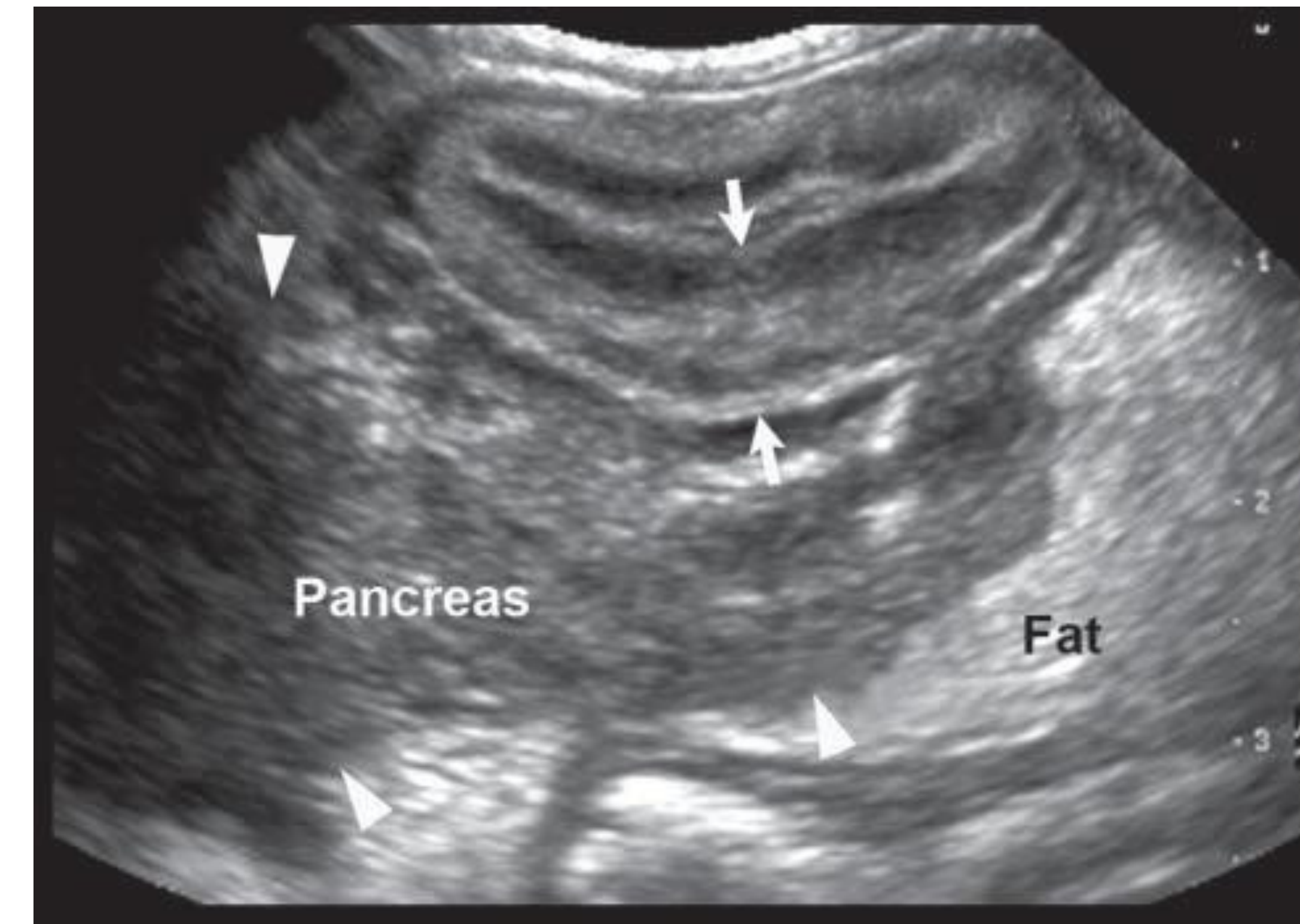


Pancreatitis + GI Involvement

XR: Sentinel loop (fixed, gas-filled duodenum)

US: Hypoechoic pancreas · Peripancreatic effusion · Rigid duodenum · CT for necrosis / complications

Duodenitis and pancreatitis in a dog



Duodenum is thickened, and the wall layering appears affected but still visible (arrows). The adjacent right limb of the pancreas is enlarged and hypoechoic (arrowheads).



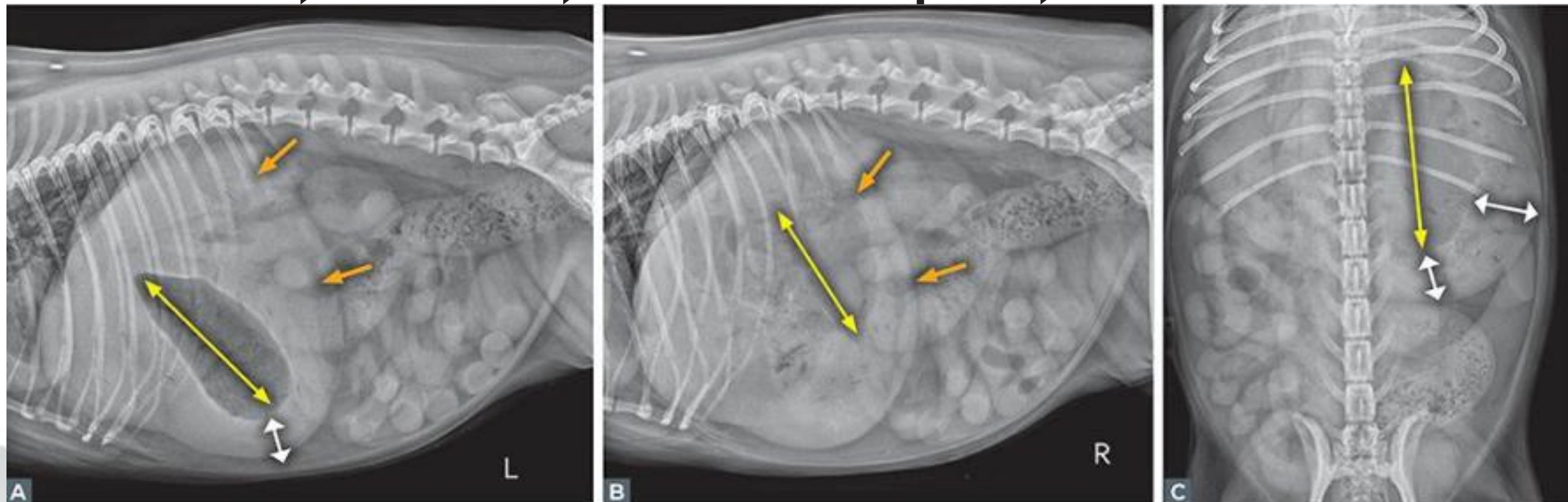
GI Neoplasia

XR: Mass effect · Displacement of bowel loops

US: Cats: diffuse SI thickening + muscularis prominence

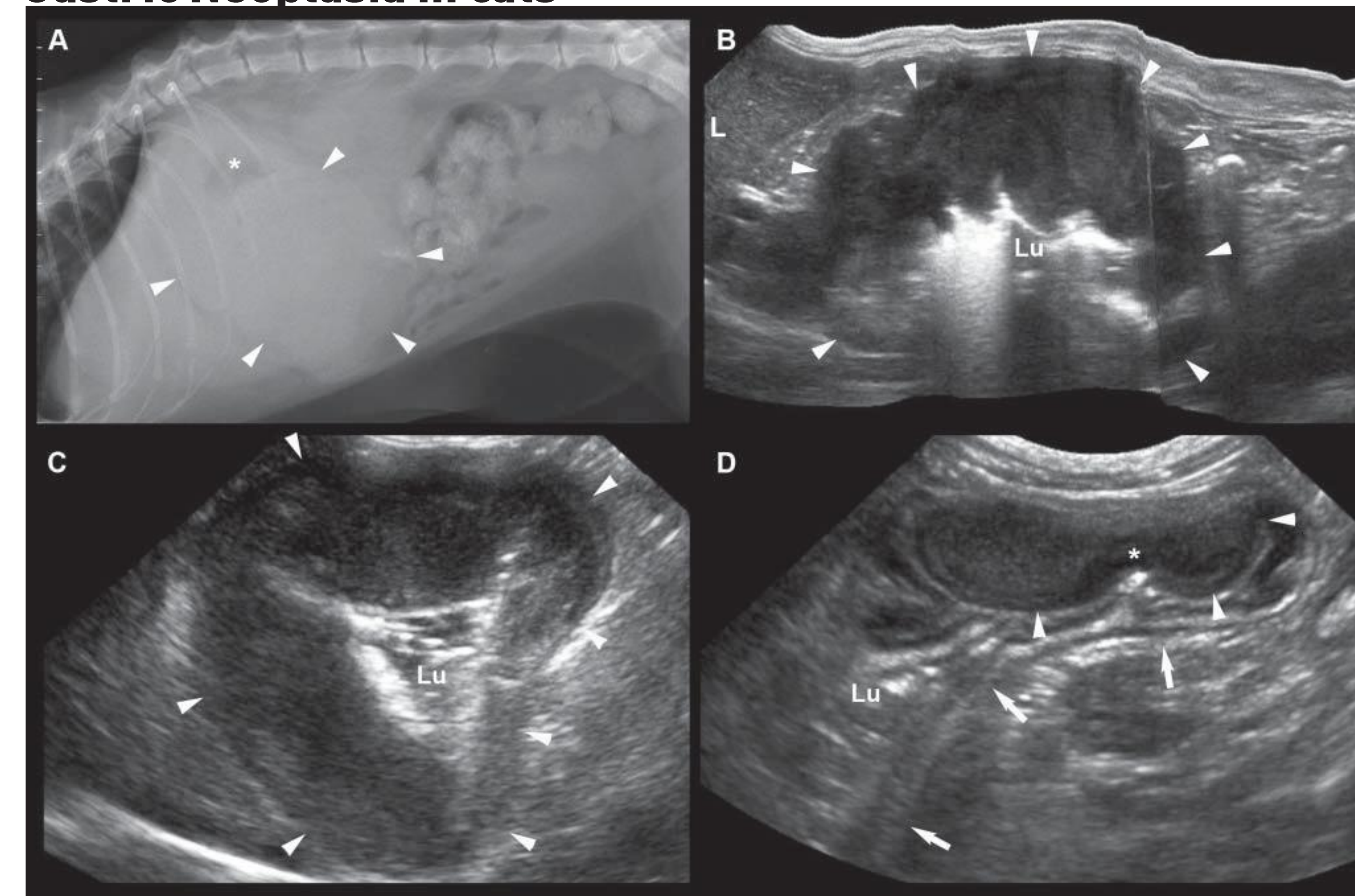
(lymphoma) · Dogs: focal mass, loss of layering · CT for staging

DDx: Gastritis, Obstruction, Gastric wall neoplasia, Pancreatitis



Pug dog that presented with retching, vomiting, and a distended abdomen. The stomach is markedly dilated due to a large amount of fluid and gas content. This presence of fluid (orange arrows) with a superimposed large gas bubble (yellow arrows) is often mistaken for a thickened gastric wall (white arrows).

Gastric Neoplasia in cats



11-year-old cat with vomiting. A large soft-tissue mass (arrowheads). B: The gastric mass (arrowheads) is poorly echogenic. C: Transverse sonogram of the same stomach shows the transmural, circumferential neoplastic infiltrate. D: 5-year-old cat, the gastric wall is thickened with loss of layers (arrowheads).



SEGMENT 4

Fibre-Related GI Conditions

MAIN FOCUS

Constipation & Obstipation · Megacolon · Trichobezoar · Phytobezoar · Fibre & IBD



Constipation & Obstipation



Radiographic Findings

- Massive colonic distension — descending colon to rectum
- Uniformly dense, compacted faecal content
- Colon diameter $> 1.5\times$ width of L5 → significant dilation
- Displacement of adjacent abdominal organs
- Pelvic inlet may be entirely occupied in severe cases

Ultrasound

- Acoustic shadowing from gas within dense faecal mass
- Wall layers may remain normal in simple constipation (vs. megacolon)

Red Flags on Imaging

- Pelvic malunion
- Extraluminal compressive masses (prostate, perineal tumour)
- Sacral / spinal lesions

Imaging Pearl: Measure colon diameter at L5 at every presentation — establish a quantitative baseline to monitor therapy response



Megacolon

Irreversible colonic dilation with smooth muscle failure — predominantly feline



Aetiology

62% Idiopathic

23% Mechanical obstruction

6% Dyssynergia

9% Other

Imaging Findings

Radiograph:

Colon uniformly distended throughout entire length · Diameter consistently > 1.5–2× L5 width · Heterogeneous faecal content · Look for pelvic deformity (mechanical cause)

Ultrasound:

Wall thickened (early) or atrophied/thinned (chronic) · Peristaltic contractions absent or severely reduced · Real-time hypomotility distinguishes from obstruction

CT:

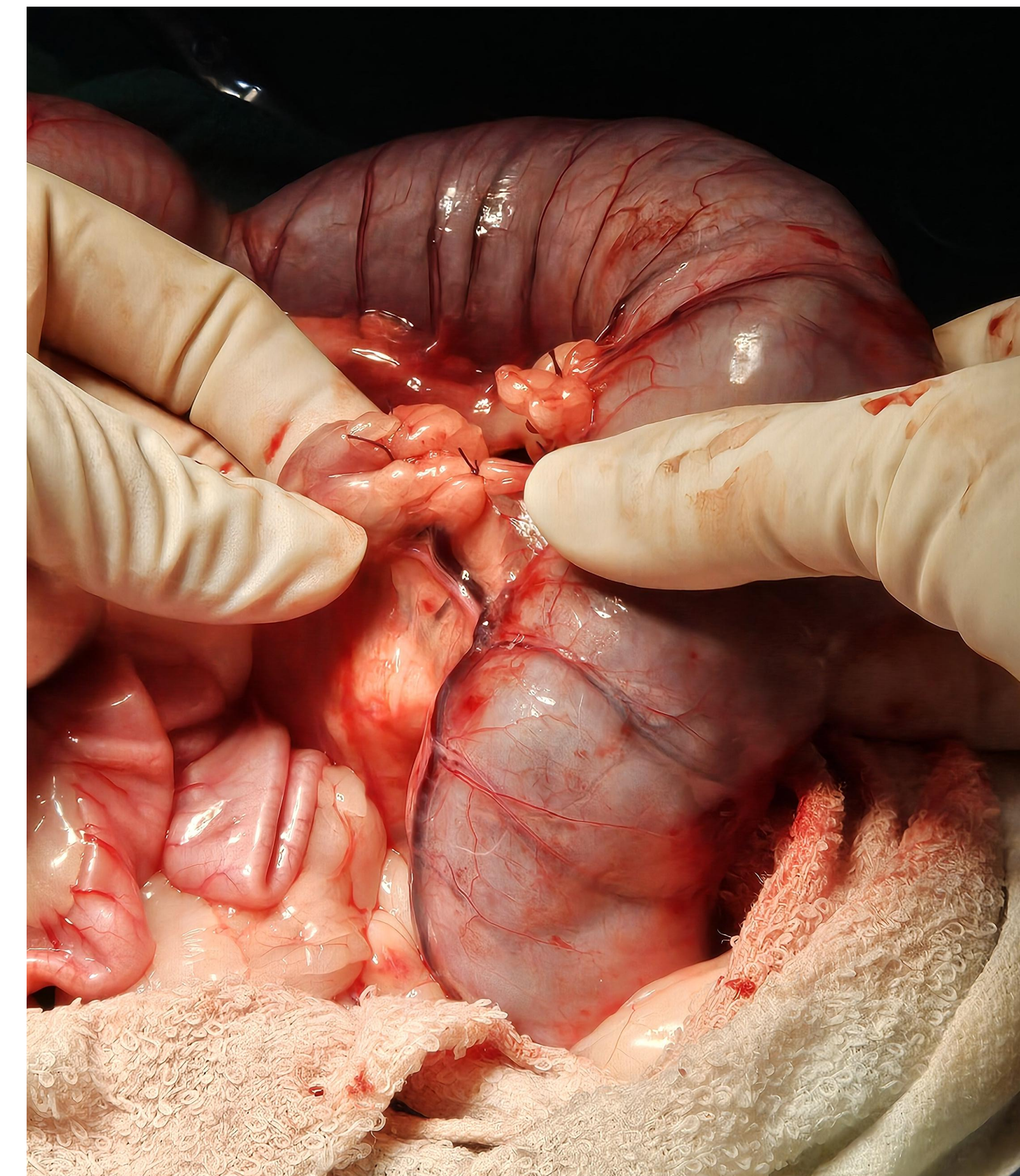
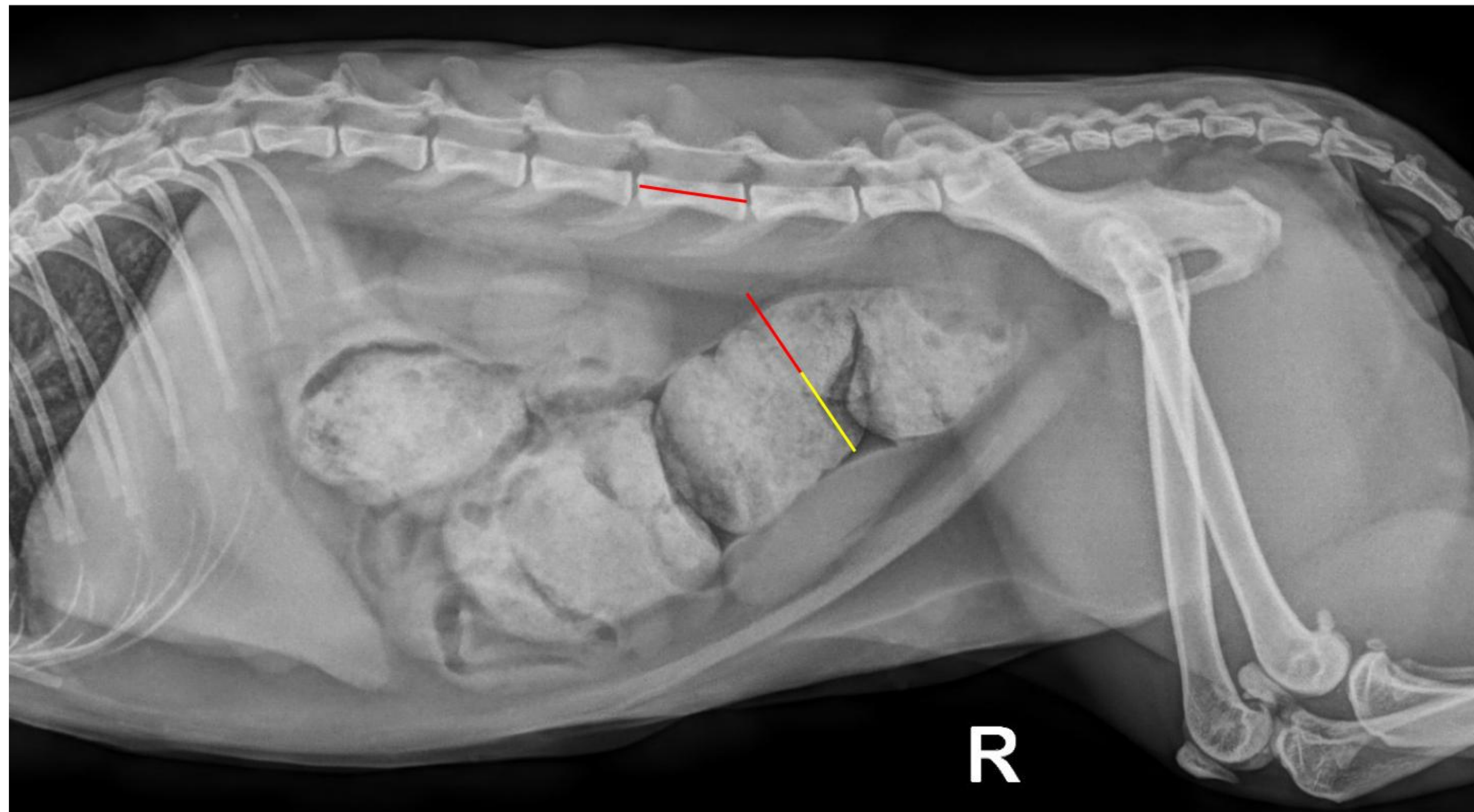
Confirms extent · Evaluates underlying obstructive lesion · Pre-surgical planning for colectomy

Key distinction: absent peristalsis on real-time ultrasound → megacolon / ileus | preserved peristalsis proximal to lesion → mechanical obstruction



Case 1 : Megacolon

- Kucing, Domestik, 7 tahun, Jantan steril
- Anamnesa : Tidak pup selama 5 hari.
- Memiliki riwayat klinis pernah colotomy untuk mengeluarkan pup. Konstipasi berulang.
- Diameter colon > 1.48 kali dari panjang lumbar 5 → mengindikasikan megacolon



Trichobezoar & Hairball-Related Obstruction



Who is at risk?

- Cats (especially long-hair breeds: Persian, Norwegian, Maine Coon)
- Dogs with excessive grooming behaviour or coat-chewing
- Multi-cat households — fur ingestion from others

Management

- Endoscopic retrieval — first-line if gastric
- Surgical gastrotomy / enterotomy — if endoscopy fails or SI location
- Prevention: Royal Canin Hairball® + regular grooming/brushing

Imaging Findings

Radiograph:

Soft tissue density mass in gastric lumen · 'Mottled' intraluminal pattern (gas displaced by hair) · Proximal small intestinal dilation if obstruction · May be isodens and MISSED on plain film

Ultrasound:

Hyperechoic intraluminal mass · Strong posterior acoustic shadowing · No internal vascularity on Doppler — KEY to distinguish from neoplasia

Barium series:

Filling defect with irregular margins within stomach · Still useful if ultrasound inconclusive

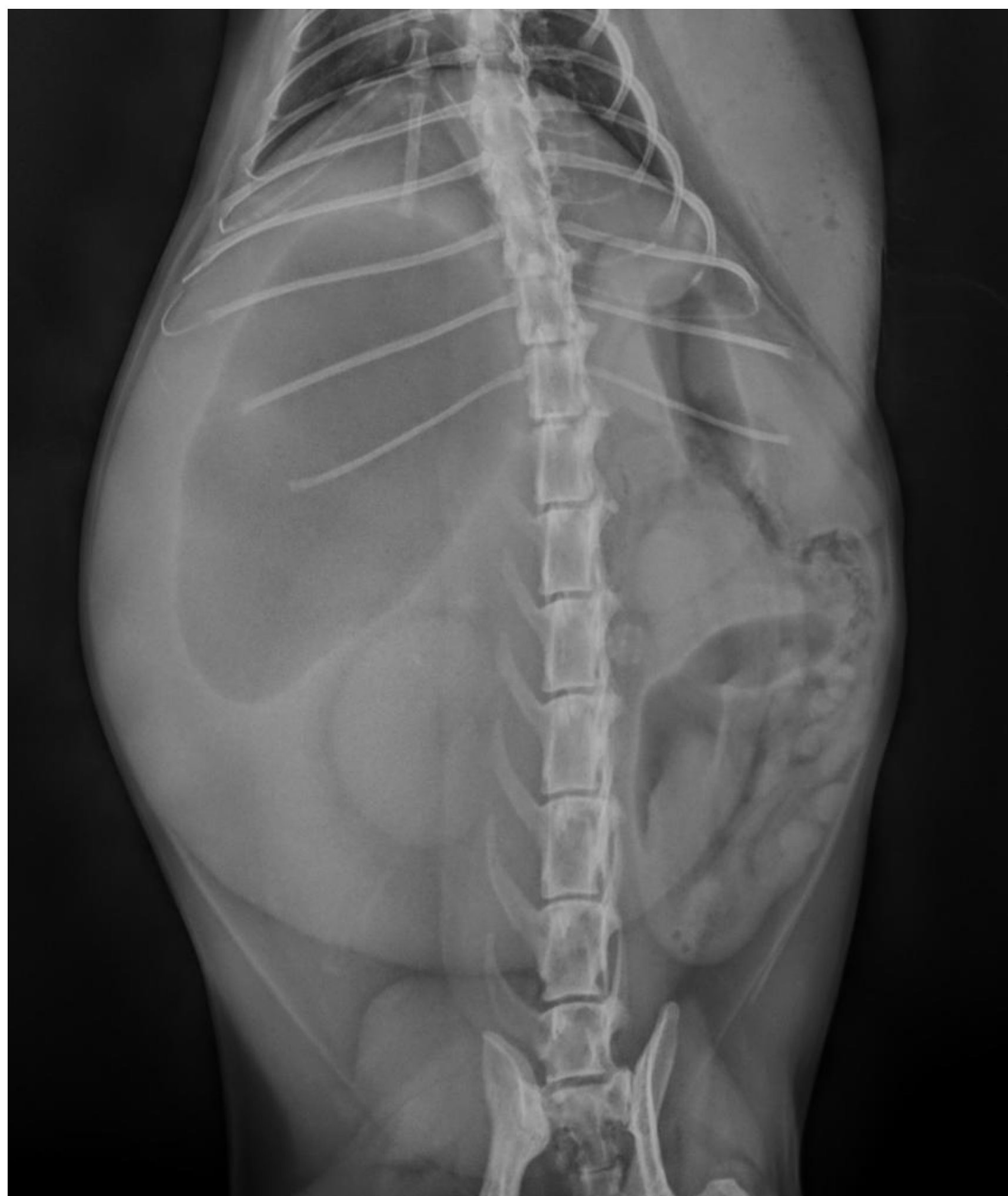
Endoscopy:

Definitive — direct visualisation of compacted hair mass · Enables retrieval if gastric location

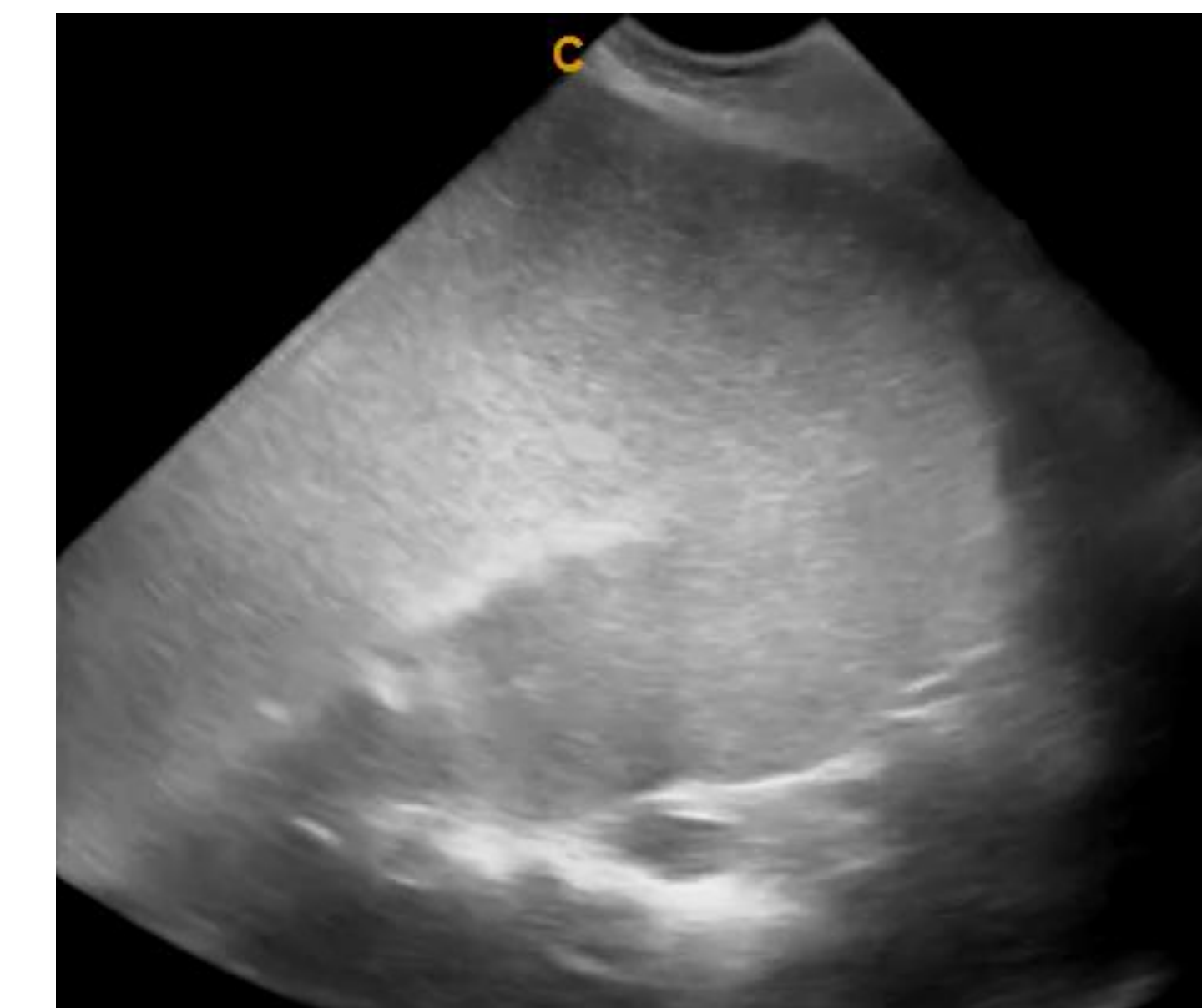


Case 2: Trichobezoar & Hairball-Related → Foreign body Obstruction

- Kucing, Persia, 9 tahun, Betina
- Anamnesa : Vomiting, perut membesar, tidak pup selama 4 hari, nafsu makan menurun tapi masih bisa disuap
- Temuan klinis : Perut tense dan membesar, lethargic, pembuluh darah colapse



- Dilatasi lambung
- Proximal small intestine dilatation dan dislokasi intestinal



Phytobezoar & Dietary Fibre / IBD

4D — Phytobezoar

Plant material (corn husks, seeds, fibrous vegetables) — more common in dogs

Radiograph:

Gas trapped within plant fibres → heterogeneous, mottled density ·
Downstream bowel dilation if obstructing · Not reliably radiopaque

Ultrasound:

Hyperechoic irregular intraluminal mass · Variable shadowing (gas entrapment) · Corrugation sign if obstruction present

Mx:

Surgical removal in most established cases · Dietary counselling +
access restriction for prevention

4E — Dietary Fibre (Royal Canin Fibre Response®) & IBD

Fibre modulates the microbiome → mucosal immunity → chronic GI inflammation

Ultrasound:

Diffuse SI wall thickening (duodenum + jejunum) · Partial/complete loss of 5-layer architecture in severe disease · Regional lymphadenopathy

Cat-specific:

Muscularis propria > submucosa thickness → HIGH specificity for small cell lymphoma → full-thickness biopsy required (not endoscopic)

Endoscopy:

Mucosal erythema, granularity, friability · Gold standard for histological diagnosis · Ultrasound guides which segment to biopsy

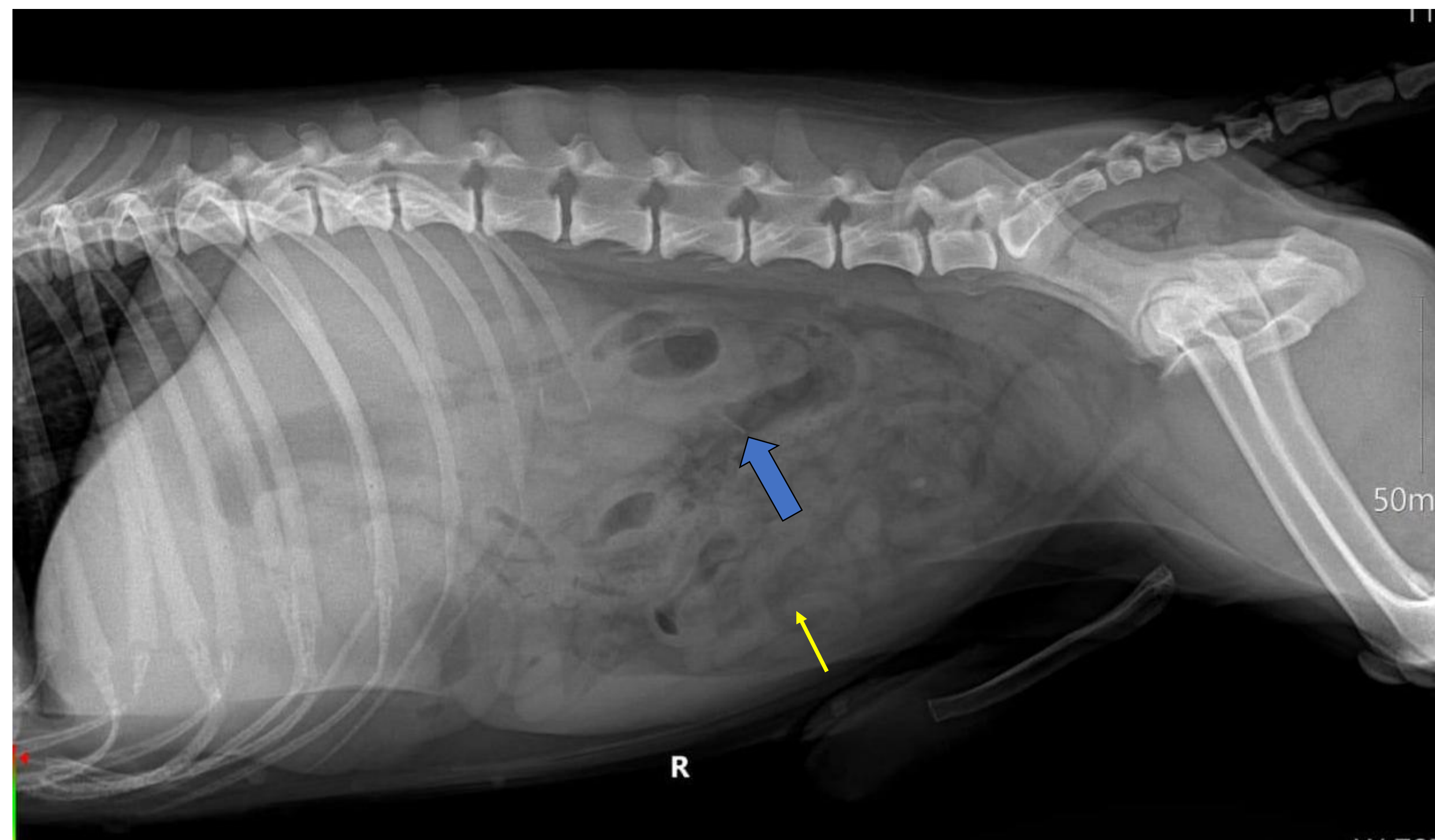
Normal mucosa on endoscopy does NOT exclude transmural disease — always correlate with ultrasound

ROYAL CANIN

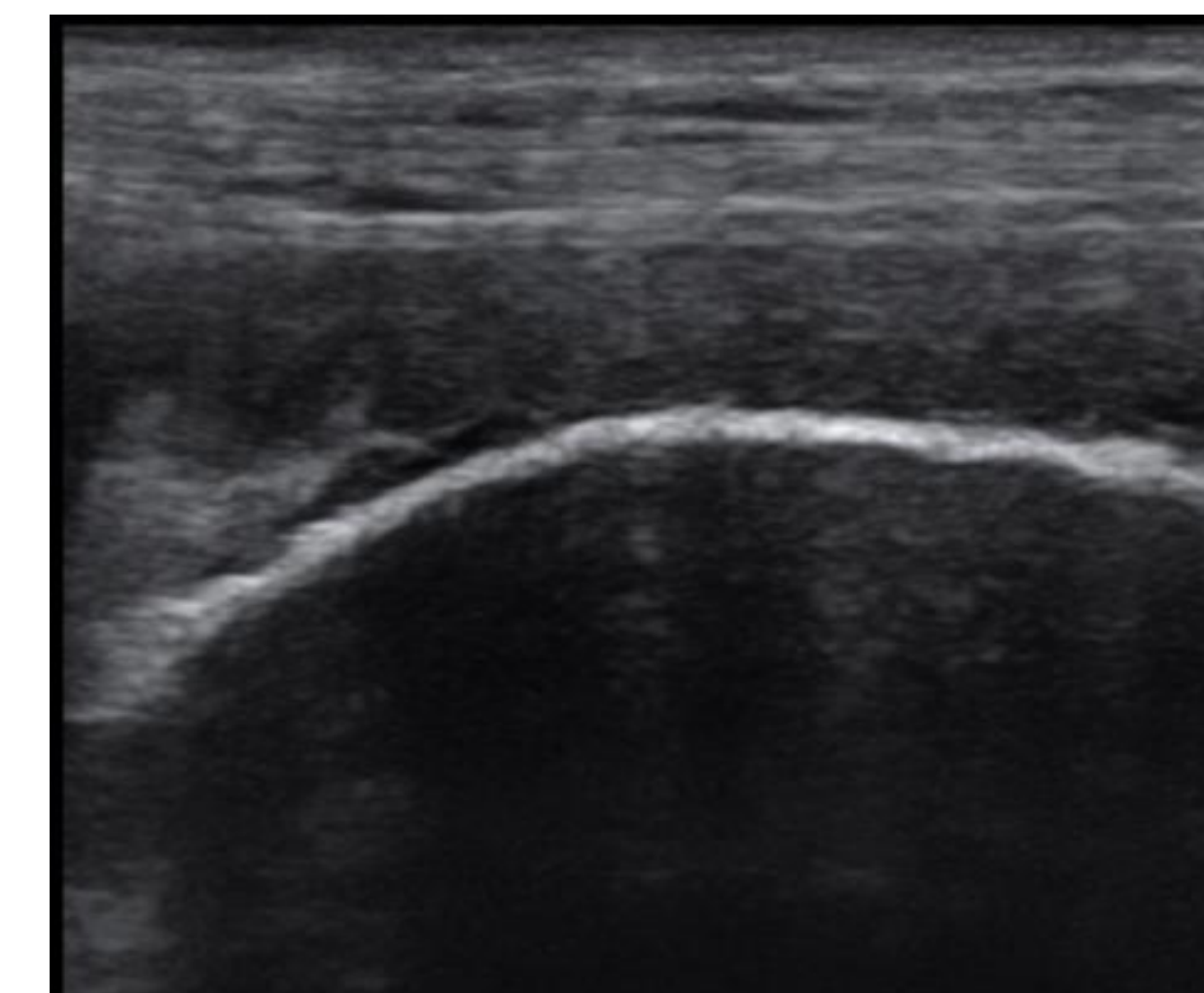
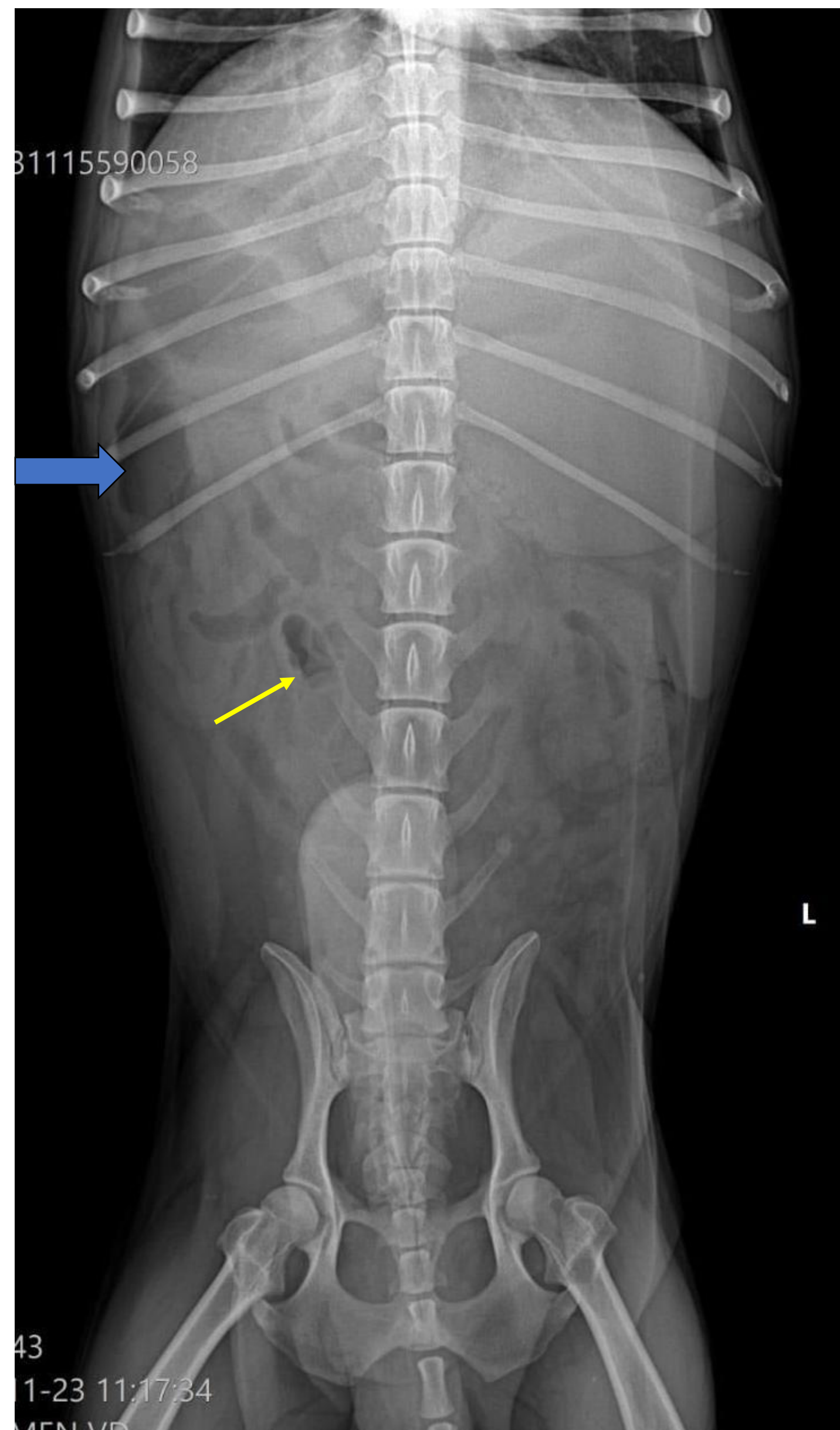


Case 3: Phytobezoar

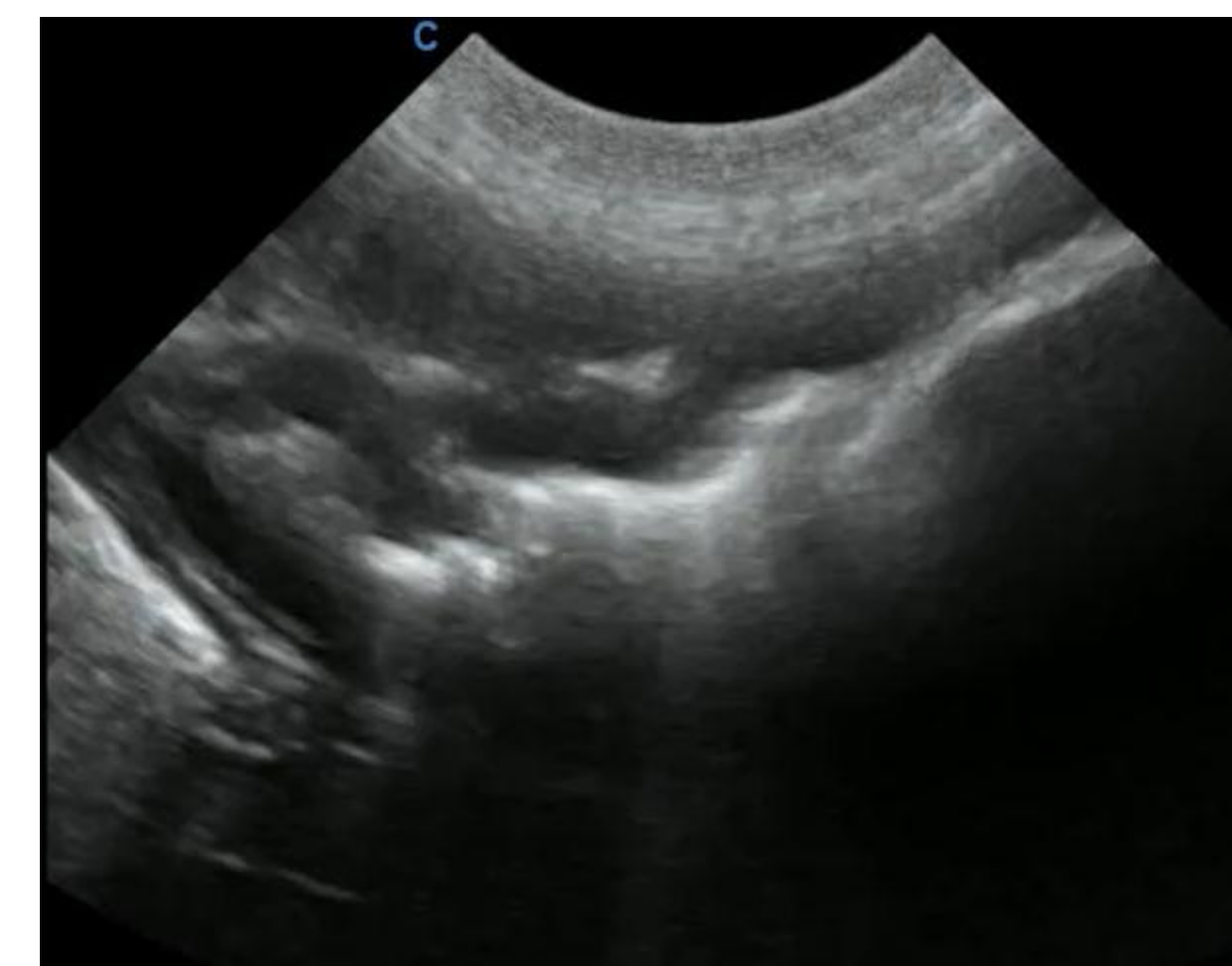
- Anjing, Beagle, 12 tahun
- Jenis kelamin : Jantan intact
- Anamnesa & temuan klinis: drooling, hypersalivasi, muntah air persisten, letargi, diare, melena



Obstruction: Two population of small intestines (normal vs dilatation)



Massa dengan acoustic shadowing pada duodenum



Gerakan antiperistaltik pada pylorus



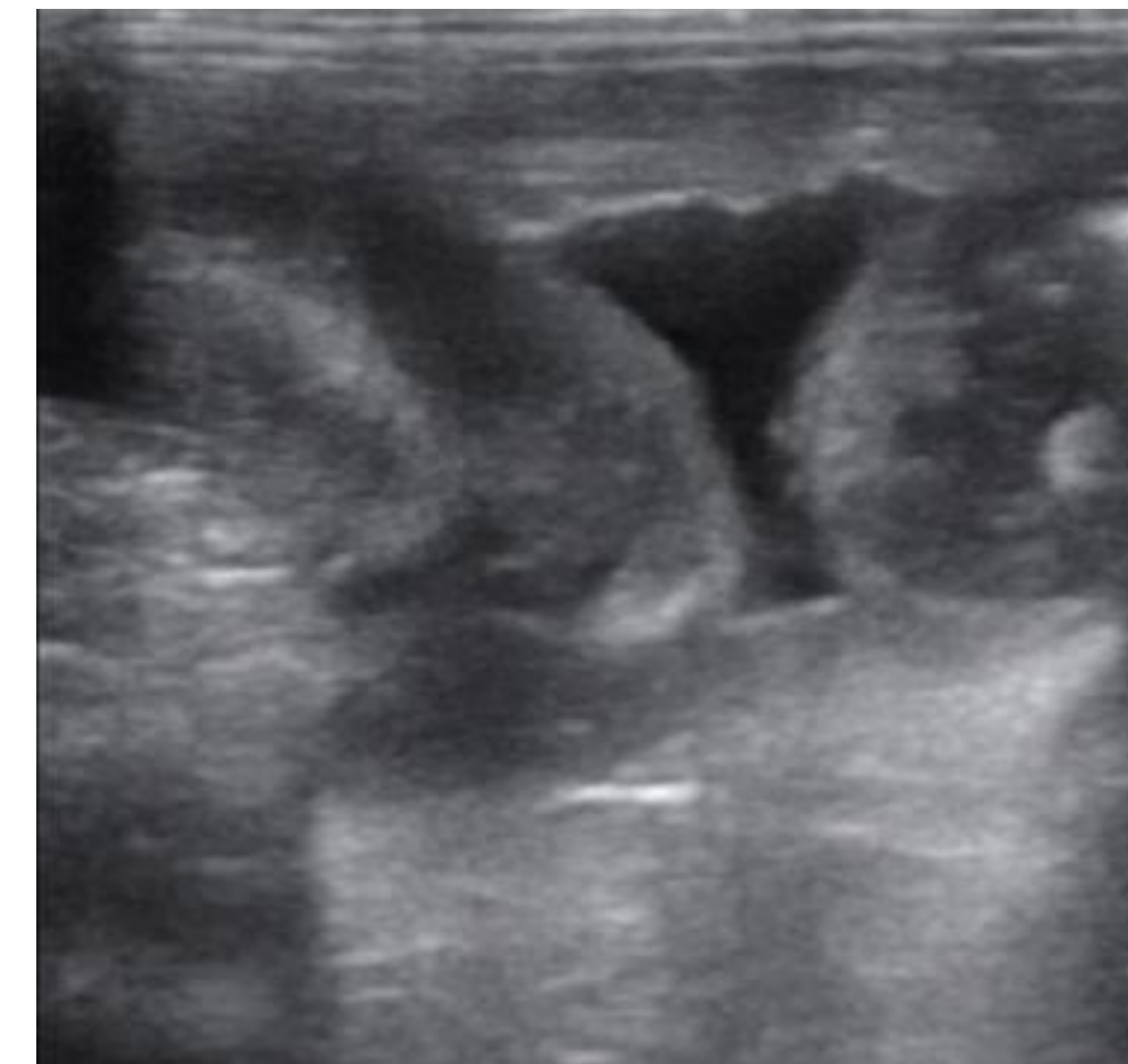
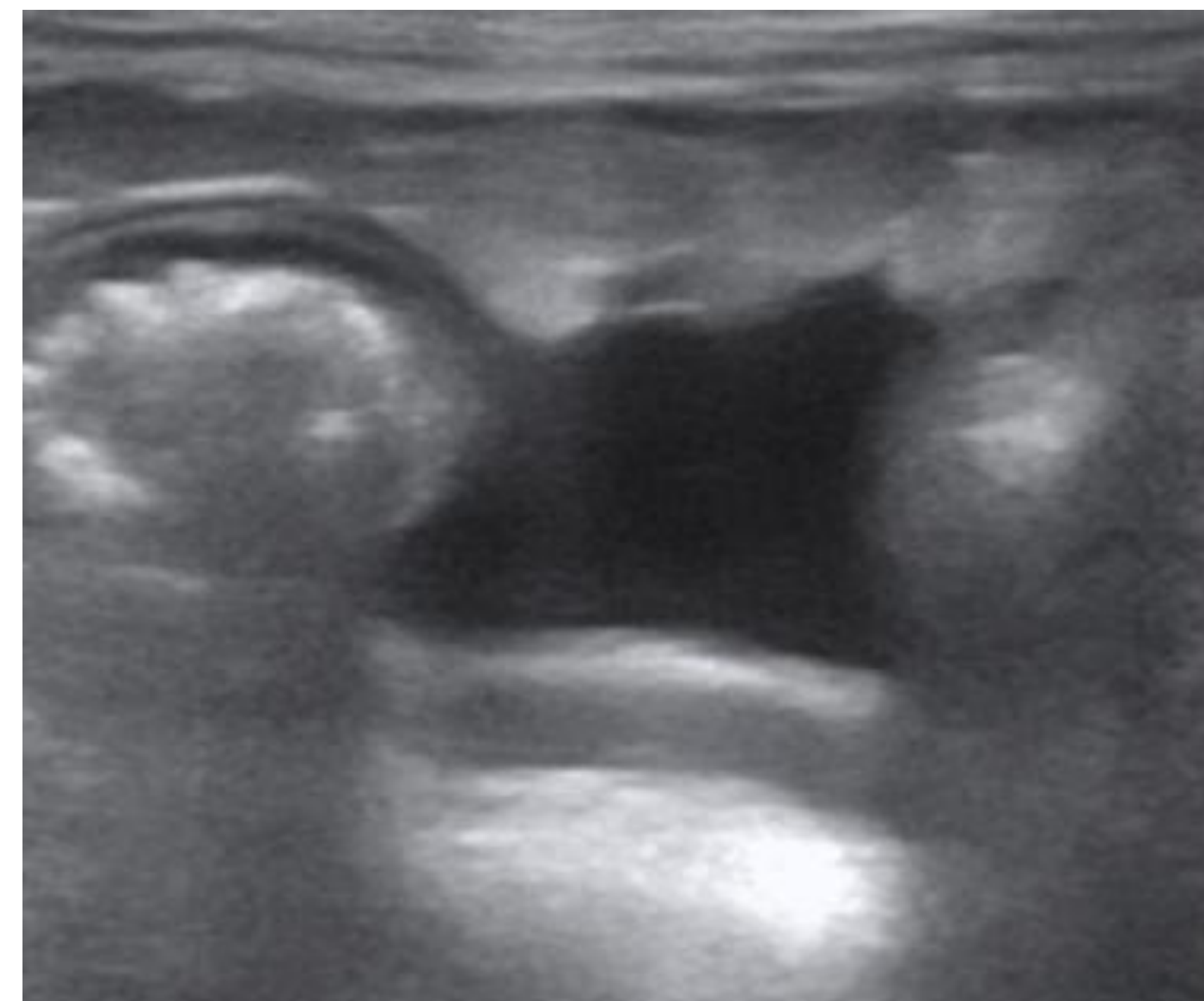
Celiotomy → biji mangga



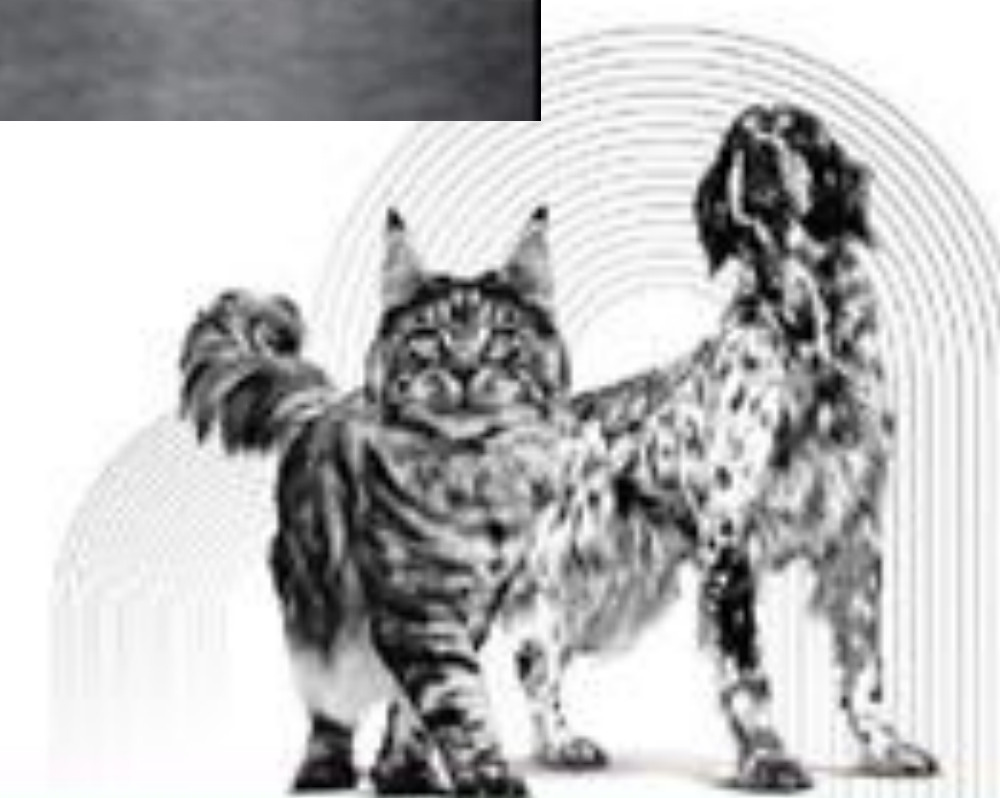
Case 4: Diare Fibre/Inflammatory Bowel Disease



- Anjing, Corgi, 10 tahun, Betina
- Anamnesa : Unappetite, muntah, diare, lemas, perut membesar
- Temuan klinis : Terasa undulasi saat palpasi, hypoalbuminemia



- “Speckles” pada mukosa usus halus
- Penebalan dinding usus halus (> 5 mm)
- Akumulasi cairan anechoic di rongga abdomen



5 Key Principles to Take Back to Practice

- 1 Ultrasound is the PRIMARY modality for GI evaluation — not just a follow-up to radiography
Real-time wall layering, motility, and Doppler are irreplaceable
- 2 Fibre-related conditions are diagnosable with radiography + ultrasound alone
CT is rarely needed for megacolon, bezoar, or constipation
- 3 Measure the colon at every feline constipation presentation
Quantitative baseline enables objective monitoring and early recognition of megacolon
- 4 'Target sign' = intussusception until proven otherwise — assess Doppler immediately
Flow present → viability → may reduce | No flow → resection mandatory
- 5 Endoscopy + biopsy is mandatory when IBD or GI neoplasia is the differential
Normal mucosa on endoscopy does NOT exclude transmural disease — correlate with ultrasound



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References

